



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Italgapura, Rajankunte, Yelahanka, Bengaluru – 560064



CALMTRIP
**An Intelligent Mobile Application for Digital travel
documentation and Trip information management**

A PROJECT REPORT

Submitted by

Aneesa MA- 20221CSE0652

Bhavana C- 20221CSE0658

Bodhini DB-20221CSE0660

Under the guidance of,

Ms. Anu Joseph

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING,

PRESIDENCY UNIVERSITY

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APRIL 2026



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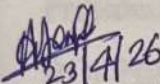
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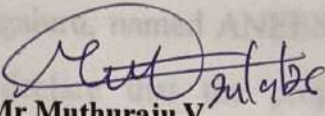


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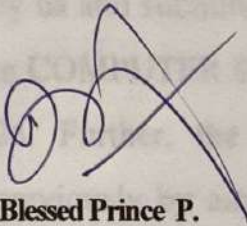
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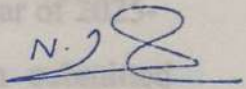
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Ms. Anu Joseph
Project Guide
PSCS
Presidency University

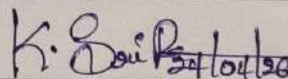
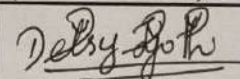

Mr. Muthuraju V
Program Project Coordinator
PSCS
Presidency University


Dr. Sampath A K
School Project Coordinators
PSCS
Presidency University


Dr. Blessed Prince P.
Head of the Department PSCS
Presidency University


Dr. N. Duraipandian
Dean PSCS
Presidency University

Examiners

Sl. no.	Name	Signature	Date
1.	Katta Sri Ram		24/04/2026
2.	Ms. DELSY JYOTHI		24/04/2026.

DATE: 26-04-2026



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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

We, the students of final year B.Tech in Computer Science Engineering at Presidency University, Bengaluru, named ANEESA M A, BHAVANA C, and BODHINI D B, hereby declare that the project work titled "CalmTrip: Development of a Mobile Travel Application for Capturing and Managing Trip-Related Information Anytime, Anywhere" has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

Aneesa MA -20221CSE0652

Bhavana C - 20221CSE0658

Bodhini DB - 20221CSE0660

Aneesa MA

C. Bhavana

Bodhini DB

PLACE: BENGALURU

DATE: 26-04-2026

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For completing this project work, We/I have received the support and the guidance from many people whom I would like to mention with deep sense of gratitude and indebtedness. We extend our gratitude to our beloved **Chancellor, Pro-Vice Chancellor, and Registrar** for their support and encouragement in completion of the project.

I would like to sincerely thank my internal guide **Ms Anu Joseph, Assistant Professor**, Presidency School of Computer Science and Engineering, Presidency University, for her moral support, motivation, timely guidance and encouragement provided to us during the period of our project work.

I am also thankful to **Dr. Blessed Prince P. Professor, Head of the Department**, Presidency School of Computer Science and Engineering, Presidency University, for his mentorship and encouragement.

We express our cordial thanks to **Dr. Duraipandian N**, Dean PSCS & PSIS and the Management of Presidency University for providing the required facilities and intellectually stimulating environment that aided in the completion of my project work.

We are grateful to **Dr. Sampath A K, and Dr. Geetha A**, PSCS School Project Coordinators, **Mr. Muthuraju V, Program Project Coordinator**, Presidency School of Computer Science and Engineering, for facilitating problem statements, coordinating reviews, monitoring progress, and providing their valuable support and guidance.

We are also grateful to Teaching and Non-Teaching staff of Presidency School of Computer Science and Engineering and also staff from other departments who have extended their valuable help and cooperation.

Aneesa MA -20221CSE0652

Aneesa MA

Bhavana C - 20221CSE0658

Bhavana C

Bodhini DB - 20221CSE0660

Bodhini DB



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Bodhini DB - 20221CSE0660

ABSTRACT

The project titled “CalmTrip: Development of a Mobile Travel Application for Capturing and Managing Trip-Related Information Anytime, Anywhere” is about transforming how people organize, document and share their journeys with an intelligent, cloud-based mobile application. Knowing that most travel apps are manual-heavy, provide limited customization and no smart automation, the startup addresses this by combining AI with location-based services to provide a seamless travel management experience.

CalmTrip is a mobile-first service that captures trip-data automatically using GPS location and geospatial analysis. LooKee senses your trips in real-time and optimally records important travel data such as routes, distance, time, and destinations with minimal user interaction. And, it allows AI-enabled trip planning — when your friend out of the blue says they’re planning a trip somewhere and the app can puke out an itinerary of cool checkpoints calibrated to time and context.

One of the system’s signature traits is its automated travel journal creation. Using your trip metadata—origin/destination, distance, time, checkpoints and photos taken—our app generates unique, first-person travel stories. It lets you store travel moments without you having to blog. The platform further provides intelligent nearby place discovery by identifying points of interest in the vicinity of the user’s position and displaying snippet-sized, context-aware tips to support exploration in transit.

It takes a privacy-forward cloud approach. Granular privacy settings allow you to switch on and off GPS tracking, photo geotagging and data-sharing settings. Secure authentication and database-level access control keeps user data isolated. It also utilizes battery-optimized GPS tracking techniques to save resources without sacrificing accuracy.

Through the combination of AI and rule-based geospatial algorithms, CalmTrip offers a roadmap for how travel planning, trip recording and journaling can be automated in a reliable, scalable manner. Overall, the project illustrates what is possible when you leverage cutting-edge mobile and AI technology to anytime, anywhere capture and control trip information and delivers a smart, secure and human-centric travel diary solution that can be deployed in the wild.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CAGR	Compound Annual Growth Rate
CDN	Content Delivery Network
CPU	Central Processing Unit
CSS	Cascading Style Sheets
ER	Entity-Relationship
GDPR	General Data Protection Regulation
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
JSON	JavaScript Object Notation
JSONB	JSON Binary (PostgreSQL data type)
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
OSM	OpenStreetMap
PDPB	Personal Data Protection Bill (India)
POI	Point of Interest
RLS	Row-Level Security

Abbreviation	Full Form
SDK	Software Development Kit
SDG	Sustainable Development Goal
UI	User Interface
URL	Uniform Resource Locator
UUID	Universally Unique Identifier
UX	User Experience
WTTC	World Travel and Tourism Council

Chapter 1

Introduction

1.1 Background

Travel has always held a sacred and non-substitutable role in the human condition. Be it for pleasure, cultural, business or spiritual reasons, travelling from one location to another has defined civilizations, expanded personal perspectives and created lasting memories that individuals cherish forever. In antiquity, travellers carved their routes in stone and papyrus, during the early modern period explorers and merchants maintained handwritten journals, while in the twentieth century, photographs and postcards reigned supreme. Each time period developed its own equipment to feed the exact same human urge: to document, memorialize and broadcast the adventure of journeying.

The explosion of smartphones in the last decade and a half completely disrupted the relationship between travel and technology. By 2024, the typical Indian city traveller has in their pocket a device more capable than the computers that mapped out the initial moon mission — GPS, a quality camera, always-on connectivity, and access to a worldwide travel app ecosystem. The promise of technology to render travel planning frictionless and travel documentation effortless has become nearly universal. But despite the incredible sophistication of today's smartphones, and the billions of dollars in travel tech investment, there's a key missing link — the experience of actually capturing a meaningful record of a trip is still mostly manual, laborious, and isolated.

Think of the average urban commuter in Bangalore nowadays. They might use Google Maps, Zomato or Swiggy to find a restaurant, BookMyShow to book an experience, and Instagram to share photos — but none of these tools talk to each other, none of them remember where the traveller has been already, and not one produces a coherent, personalized record of the day's journey. The traveller comes home to a camera roll and a Google Maps timeline and maybe a mental impression of what it was like — but no story, no structured record, no automatically generated diary entry that captures the texture of the day. The documentation gap is not about what people want — surveys time and again indicate that travellers want to document. It's a question of friction – the work it takes to write journal entries by hand, tag

locations and stitch together a cohesive narrative is just not commensurate with the impulsive, immersive nature of travel itself.

This issue, though small in scale, has major repercussions. Travel writing is closely related to mental health — positive psychology has demonstrated the tight connection between reflective journaling and positive memory formation, with documented benefits for mood, stress, and happiness. Research in the travel domain in particular has found that travellers who track their journeys experience more satisfaction with their travels and feel more connected to the locations they visit. Technology's failure to properly support this documentation, then, is not just a product design defect — it's a tangible deficit in human flourishing.

On the scheduling side, the difficulty is just as severe. The typical Indian city consumer invests more than three and a half hours planning one mini-trip, hopping through a fractured ecosystem of review sites, map apps, feeds and friends. All existing trip planning tools — from TripAdvisor to Google Travel to Yelp — work on a fundamentally impersonal model: they provide popularity-ranked suggestions tailored to the generic user, with zero context about what the person has already experienced, where they like to go, how far they'll walk, or the subtle flavor of experience they're seeking on a particular day. A tourist who just wants a lazy Sunday afternoon wandering Basavanagudi's heritage cafés has no tool that hears this, knows their past travel experiences, and creates a personalised, strollable plan in natural language.

The arrival of LLMs is the first truly transformative development in this space since the smartphone itself. Models like Google Gemini, OpenAI GPT-4, and Anthropic Claude have shown the incredible ability to comprehend natural language prompts, reason about constraints and preferences, and produce structured, contextually appropriate output across a massive variety of domains. Applied to travel planning, this ability implies that a user can just say 'I want a quiet two hour walk through old Bangalore with good coffee stops' and get a customised, actionable itinerary — without completing forms, ticking categories, or inputting parameters. Applied to travel documentation, this same facility implies that the rich metadata of a completed trip — start location, end location, checkpoints along the way, distance, photos — can be automatically converted into a coherent, personalized journal entry, in the traveller's voice.

But the convergence of AI with location-based services presents a deep accountability that most of the present industry has been reluctant to address. Location data is one of the most sensitive types of personal information a mobile device can gather. A stream of GPS coordinates is more than geographic data — it's a map of an individual's life. It exposes their address, place of work, churches, hospitals, social circles – and a hundred other private information that the individual hasn't opted to share. The prevailing model in location-aware mobile apps has been to regard this information as merely a raw material to be mined: coordinates are uploaded to cloud servers in real time, stored indefinitely, and leveraged for advertising targeting, market research and a host of secondary uses that users neither comprehend nor consent to.

The worldwide regulatory response to this paradigm has been unequivocal. Even the EU's GDPR has only put strict requirements on how to collect, process and keep personal data since 2018. California CCPA soon afterwards and in 2020. Most notably for India, the DPDP Act was passed by Indian Parliament in August 2023, introducing for the first time a robust legal framework for digital personal data protection in India. It also mandates clear, purpose-specific consent when collecting personal data such as location data, provides the right to erasure, imposes significant penalties for breaches. This law has turned privacy from a competitive differentiator into a legal necessity for every app in the Indian market.

CalmTrip was born at exactly this convergence of four forces — the inherently human desire to document and plan travel experiences; the unique power of generative AI to intelligently automate these tasks; the paramount need to protect location data privacy; and the Indian domestic travel market, the fastest-growing travel market globally, yet the most underserved by AI-powered tools. We named it CalmTrip because that's what we designed it to be — your travel companion that stays quietly in the background, needs no attention or manual input, respects your privacy absolutely, and delivers smart, personalised help without drama or friction. The app automatically recognizes when a trip starts, follows along with a battery-efficient GPS algorithm, creates a detailed travel diary entry when the trip is over, and provides personalized suggestions for the next destination — all the while storing every bit of sensitive location information on the user's phone unless they decide to share it.

The technical underpinning of CalmTrip lies in a thoughtfully engineered hybrid architecture that integrates the generative capabilities of Google's Gemini LLM with the accuracy and dependability of deterministic geospatial algorithms. Natural language trip planning, journal

generation, and point-of-interest ranking are powered by the Gemini API, which shines at open-ended, contextually rich tasks where human-quality reasoning and creative expression are needed. GPS trip detection, distance calculation using the Haversine formula, and statistical travel analytics are solved by rule-based algorithms locally on your device, providing the accuracy, determinism, and offline reliability safety-critical measurements tasks require. This split of labor is not accidental: it corresponds to a principled understanding of where AI provides distinct value and where rule-based computation is better suited, producing a system both smarter and more dependable than either approach on its own could deliver.

1.2 Statistics of the Project Domain

\$9.5T	700M+	68%	87%
Global travel & tourism GDP (2023, WTTC)	Active travel app users worldwide (2024)	Travellers dissatisfied with manual logging	Users concerned about GPS data privacy

The world’s travel & tourism economy generated USD 9.5 trillion to world GDP in 2023, accounting for 9.1% of global economic output. International tourist arrivals rose to 1.4 billion, which represents 88% of pre-pandemic levels. Overall, the online travel market alone was worth more than USD 1.06 trillion, and is growing at 11.4% CAGR through 2030. In India, domestic tourism had more than 2.3 billion visits in 2022–23 and the travel app market is expected to cross USD 26 billion by 2027.

More than 700 million people use travel apps globally (Statista, 2024). A Google Travel Insights survey discovered 73% of travellers rely on mobile apps while on the road, but post-trip record-keeping continues to be severely underserved — 68% of frequent travellers are frustrated by the inconvenience of manually logging trips, and 54% have given up travel journaling altogether. The typical traveller dedicates 4.2 hours planning one trip fragmented across platforms (TripAdvisor). Generative AI is set to hit USD 1.3T market by 2032; 40% new consumer apps to use LLMs by 2025 Privacy fears are high: 87% of location app users are concerned about GPS data use, and 62% have refused location permissions to at least one travel app (Pew Research, 2023).

1.3 Prior Existing Technologies

Current solutions come in five categories, each solving only a portion of the full travel planning and documentation issue.

1. General-Purpose Trip Planners (TripIt, Wanderlog, Sygic Travel): Combine bookings, recommend attractions with structured inputs and popularity lists. No natural language interface, no GPS tracking, no AI generated documentation.
2. Navigation & Location Trackers (Google Maps Timeline, Life360): Auto-log movements via continuous GPS but are tracking dashboards only — no trip planning, no narrative generation. Need always-on cloud transmission of exact location data.
3. AI Chatbots (ChatGPT, Gemini, Perplexity): Create itineraries in natural language but are stateless — no knowledge of individual travel history, real-time GPS location or live geospatial API. Chatbots, not unified booking platforms.
4. Travel journaling apps (e.g. Day One, Polarsteps, Roadtrippers) was able to record manually with photo uploads and map logging. No robot copy, you write every word. Top, abandonment rates due to effort.
5. POI Discovery Systems (Yelp, Foursquare, TripAdvisor) Rich databases with proximity-based discovery but no personalisation beyond aggregate ratings, no natural language interface, and no integration with a personal travel diary.

Table 1.1 — Features Comparison of existing applications with CalmTrip

Feature	Trip Planners	Trackers	AI Chatbots	Journals	POI Apps	CalmTrip
NL Trip Planning	✗	✗	✓ (stateless)	✗	✗	✓ (personalised)
Auto GP S Detection	✗	✓	✗	Partial	✗	✓
AI Journal Generation	✗	✗	✗	✗	✗	✓
Personalised to History	✗	✗	✗	✗	✗	✓

Feature	Trip Planners	Trackers	AI Chatbots	Journals	POI Apps	CalmTrip
AI POI Discovery	Partial	✗	✓ (generic)	✗	✓	✓ (AI-ranked)
Local-First Privacy	✗	✗	N/A	Partial	✗	✓

1.4 Proposed Approach

CalmTrip offers a hybrid AI and geospatial computing architecture based on three tenets — intelligence, integration and privacy by design.

Hybrid AI–Rule-Based Architecture: Google Gemini LLM manages creative, open-ended activity — trip planning, journaling, POI ranking. Rule-based deterministic algorithms cover GPS detection, distance, and analytics, guaranteeing accuracy, auditability, and predictable performance.

Local-first, privacy-by-design data model — All GPS data, trips, and photos are stored on-device first. Cloud sync through Supabase is an explicit user action. Row-Level Security policies provide per-user data isolation at the database layer. Five standalone privacy controls provide users with detailed control over their information.

Context Aware AI Planning with Personal History Instead of stateless chatbots, CalmTrip injects a summary of the user’s actual past trips — destinations, distances, patterns — into every Gemini planning prompt, generating itineraries that avoid recently visited locations and adhere to declared time budgets.

Battery-Optimised Threshold-Based GPS Detection: GPS is turned on smartly only when there is motion beyond 100 m; readings worse than 50 m accuracy are ignored; a 60-second inactivity timeout automatically ends sessions. This wastes about 60–70% less battery than constant tracking.

Progressive Geocoding Pipeline: AI-generated place names are geocoded through three successive levels — full name + city + country, + name + country, + truncated first token + city + country — with 85-90% success for Indian cities by the Nominatim API.

1.5 Objectives

O1 – An AI-powered natural language trip planning system that understands user intent from conversational input and automatically converts it into structured, actionable itineraries complete with geocoded checkpoints, estimated travel times, and logically sequenced stops

O2 – A smart background GPS trip detection mechanism that automatically identifies when a trip has started or ended using movement patterns and location signals, optimised to consume minimal battery while maintaining high detection accuracy and low false positive rates

O3 – A fully automated AI travel journal generation system that synthesises trip metadata such as locations visited, duration, and route taken into rich, human-readable narrative paragraphs — requiring zero manual effort from the user at any stage

O4 – An intelligent nearby place discovery feature that scans a 600-metre radius around the user's current location in real time, retrieves relevant points of interest, and uses AI to rank and describe them based on contextual relevance and user preferences

O5 – A privacy-first, local-first data architecture that ensures all personal travel data is stored exclusively on the user's device by default, supported by a minimum of five independent, user-controlled privacy settings that give full transparency and control over data usage

O6 – A rule-based travel insights and analytics engine that accurately aggregates trip distances, identifies patterns in travel behaviour, and classifies users into meaningful traveller archetypes based on their historical activity and movement data

1.6 Sustainable Development Goals (SDGs)

SDG 8

Decent Work & Economic Growth

CalmTrip's AI-powered POI discovery directs users to local cafes, artisan markets, and heritage sites overlooked by mainstream platforms, channelling tourist footfall to small businesses and supporting micro-tourism economies. Target: 8.9 — Sustainable tourism that creates jobs and promotes local culture.

SDG 9 Industry, Innovation & Infrastructure

The hybrid AI-rule-based architecture, progressive geocoding pipeline, and battery-efficient trip detection algorithm represent original engineering contributions to the field of AI-integrated geospatial systems. Target: 9.5 — Enhance scientific research and upgrade technological capabilities.

SDG 10 Reduced Inequalities

Built for mid-range smartphones with local-first storage, CalmTrip democratises intelligent trip planning previously available only through expensive tour operators or premium subscriptions. Target: 10.2 — Promote social and economic inclusion of all.

SDG 11 Sustainable Cities & Communities

The planner generates walkable, low-impact urban itineraries. The POI engine surfaces parks, libraries, and heritage sites, encouraging engagement with the built and natural heritage of urban communities. Target: 11.4 — Protect and safeguard cultural and natural heritage.

SDG 16 Peace, Justice & Strong Institutions

CalmTrip's local-first, opt-in architecture is a direct engineering implementation of the right to digital privacy. Every design decision — local storage, RLS, granular toggles — demonstrates that AI-powered services need not require invasive data collection. Target: 16.10 — Protect fundamental freedoms.

Chapter 2

Literature Review

The creation of CalmTrip benefits from a deep well of previous research from five separate but complementary areas: automated travel planning and combinatorial optimisation, large language model capabilities and prompt engineering, GPS-based activity recognition and energy-efficient location tracking, privacy-preserving mobile system architectures, and geospatial computing algorithms. This chapter surveys the relevant work in each area, pinpointing the particular insights that influenced CalmTrip’s design and contextualizing the system’s contributions within the state of the art.

2.1 Automated Travel Planning and the Orienteering Problem

The academic study of automated travel planning originates with the Orienteering Problem (OP), a combinatorial optimisation formulation introduced to the operations research literature in the 1980s. The OP captures the problem of choosing a subset of locations from a larger set and building an ordered path through the chosen locations that maximises total reward subject to a time or distance budget. This summarizes the core of travel planning — a traveller can’t visit every possibly interesting location in a city in an afternoon, and the challenge is thus to choose the best subset and sequence given time. Vansteenwegen, Souffriau, and Van Oudheusden (2011) supply a thorough survey of OP variants that continues to serve as the field’s default reference, including the Team OP (for teams of travellers), the Time-Dependent OP (in which travel times between locations depend on time of day), and stochastic formulations that incorporate uncertainty in location rewards or travel times.

The OP is computationally challenging: the canonical formulation is NP-hard in general, and even moderate-sized instances with 20–50 locations pose a serious challenge to exact solution methods. This computational intractability has motivated much of the research into heuristics and metaheuristics. Gavalas, Konstantopoulos, Mastakas and Pantziou (2014) survey algorithmic approaches to the Tourist Trip Design Problem (TTDP), a generalisation of the OP that incorporates practical travel constraints including location opening hours, category preferences, mandatory visits and multi-day planning horizons. Their taxonomy differentiates between personalised and non-personalised planners, offline and online planning modes, and

single-day and multi-day itineraries. They find this mix of personalised, online, single-day planning — exactly the use case that CalmTrip targets — to be the most pragmatically valuable and the most algorithmically challenging, observing that the personalised aspect in particular resists efficient classical formulations because preference functions are difficult to specify, elicit, and encode into optimization objectives.

This line of work directly inspires CalmTrip’s choice to entrust checkpoint selection to a large language model rather than a classical optimisation algorithm. OP and TTDP literature also shows that, even if you had a perfectly specified objective function, planning optimal short urban itineraries is computationally difficult and yields contextually inappropriate, though technically optimal results — a sequence of locations that minimises total distance is not necessarily an enjoyable or culturally coherent walk. The Gemini model, by contrast, leverages its training on large corpora of travel writing, local reviews, and urban geography to generate plans that are contextually relevant, culturally consistent, and compellingly narrative in nature, while honoring the time lengths specified by the user. The model’s implicit knowledge of which Bangalore cafes are atmospheric vs commercial, which heritage neighbourhoods are photogenic vs inaccessible, and which combinations of stops make a satisfying afternoon is exactly the kind of qualitative knowledge that classical optimisation cannot encode.

Ricci, Rokach, and Shapira (2015) in their authoritative Recommender Systems Handbook provide a conceptual framework that applies equally well to understanding CalmTrip’s approach. They distinguish three basic recommendation patterns: collaborative filtering, which infers the user’s preferences from the behaviour of similar users, content-based filtering, which recommends things similar to those the user has already acted upon, and knowledge-based recommendation, which explicitly reasons about the user’s requirements and item properties. Travel planning, they observe, is especially amenable to knowledge-based methods as travel decisions are rare, high-stakes, and highly context-dependent — the right recommendation for a user on a rainy Sunday afternoon is fundamentally different from the right recommendation for a warm Saturday morning, and collaborative filtering across thousands of users cannot capture this granularity. CalmTrip’s planning module is effectively a knowledge-based system that supplements explicit user constraints (length, experience type) with the user’s travel history, employing Gemini as the reasoning engine that integrates these inputs into a recommendation.

2.2 Large Language Models and Prompt Engineering

The key capability enabling CalmTrip’s AI planning and journal generation modules — a large language model’s ability to comprehend nuanced natural language requests, reason about constraints, and generate structured, high-quality output — is the culmination of a research trajectory that begins with the transformer architecture (Vaswani et al., 2017) and ends, for now, with the demonstration of GPT-3’s few-shot and zero-shot generalisation capabilities (Brown et al., 2020). The sheer scale of GPT-3 — 175 billion parameters trained on around 45 terabytes of text — led to what Brown et al called emergent capabilities: competencies that smaller models lacked and emerged qualitatively rather than incrementally as model scale increased. One of these new capabilities was the ability to obey natural language instructions for structured generation tasks, such as generating valid JSON objects from a description of the desired schema — a capability that CalmTrip’s trip planning module leans on directly.

Brown et al.’s foundational discovery — that large enough LMs, in addition to learning a language, can also complete a wide variety of tasks via in-context learning without any task-specific fine-tuning — laid conceptual groundwork for prompt engineering as a discipline. Instead of training a new model for each application, developers could write meticulously engineered prompts that conditioned the model’s output via examples, direction, and context. The explosion of research into prompt engineering after GPT-3’s publication generated a fertile literature of methods that CalmTrip’s prompts tap into.

Wei, Wang, Schuurmans, et al. (2022) introduced chain-of-thought prompting, showing that including intermediate reasoning in prompts — either as few-shot examples of step-by-step problem solving, or as instructions to ‘think step by step’ — dramatically improved performance on multi-step tasks, especially arithmetic, commonsense reasoning, and symbolic manipulation. CalmTrip’s planning prompt includes aspects of this thinking by explicitly asking the model to ‘take into account the user’s constraints before generating checkpoints recommendations’: the prompt asks the model to ensure that the total suggested checkpoints fit within the stated time budget, that checkpoint names are specific enough to be geocodable, and that the itinerary feels like a coherent, enjoyable stroll rather than a random scattering of locations. This structured reasoning guidance makes its plans better and more constrained-adherent.

Ouyang, Wu, Jiang, et al. (2022) present RLHF fine-tuning as the magic sauce that makes a raw pre-trained language model into a well-behaved instruction-following assistant. As their InstructGPT paper showed, RLHF-tuned models were dramatically more aligned with user intent — better at following instructions, less likely to produce harmful content, and more honest about uncertainty — than the raw pre-trained models from which they were derived. All Gemini model family members utilized for CalmTrip have been fine-tuned with similar techniques, so CalmTrip is able to coax reliably structured JSON output through prompt engineering alone — without any application-specific fine-tuning.

Liu, Yuan, Fu, Jiang, Hayashi, and Neubig (2023) offer a detailed survey of prompting methods, classifying approaches along the axes of task type (generation, classification, reasoning), input type (natural language, code, structured data), and output type (free text, structured data, classification labels). They recognize output format specification — explicitly providing the desired JSON schema within the prompt — as one of the most powerful approaches for structured generation tasks, observing that models consistently adhere to specified schemas when the schema is well and fully defined. CalmTrip’s planning prompt employs exactly this strategy — the full JSON schema for the AiTripPlan object, including types and constraints for each field, is included in the prompt, so the model’s output can be parsed reliably without complicated post-processing.

More recent work on prompt reliability for production applications has emphasized the need for graceful degradation: designing prompts and application code so model failures — sporadic failure to output the requested format, insertion of explanatory text alongside the JSON, markdown code fences — are dealt with transparently rather than crashing the application. CalmTrip’s JSON extraction logic — that strips markdown fences and tries a few parse strategies before falling back to a retry — embodies the pragmatic advice from this corpus.

2.3 GPS-Based Activity Recognition and Energy-Efficient Tracking

The research literature on GPS-based activity recognition furnishes both the conceptual framework and the ground-truth calibration for CalmTrip’s trip detection algorithm. Lara and Labrador’s (2013) excellent survey of wearable sensor-based human activity recognition identifies GPS as the mainstay modality for large-scale movement classification tasks such as trip detection, transportation mode classification and route tracking. They observe a persistent

conclusion throughout the surveyed literature: threshold-based approaches to such coarse-grained event detection — when a user starts or ends a trip, shifts from pedestrian to vehicular movement, or arrives at an important location — continue to be competitive with machine learning classifiers in terms of accuracy while enjoying dramatically better interpretability, computational efficiency, and energy consumption properties. This finding is direct evidence why CalmTrip opted for a threshold-based state machine for trip detection instead of an on-device machine learning classifier.

Paek, Kim, and Govindan (2010) provide the seminal study of energy-efficient GPS positioning for smartphones, proposing a duty-cycled GPS receiver — alternating between sampling intervals and sleep intervals — triggered based on signals from lower-power sensors such as the accelerometer and gyroscope. Their insight was that accelerometer-based dead reckoning could be used to figure out when meaningful movement was taking place, enabling GPS activation to be concentrated in spans of actual travel instead of spread evenly over time. They observed that their implementation yielded energy savings of 40 to 60% relative to fixed-rate GPS sampling, with position accuracy degradation of under 15% — a very favorable trade-off for applications for which approximate position, rather than centimetre-level precision, is adequate. CalmTrip’s philosophy matches Paek et al.’s framework: by activating GPS only when position updates meet distance and accuracy criteria, it instead focuses GPS power consumption on the times that provide the most documentary benefit.

Zheng (2015) offers a very useful review of trajectory data mining that is relevant to CalmTrip’s trip detection algorithm design. The survey addresses methods for trip segmentation — defining the limits between individual travel episodes in a continuous GPS trace — and stop detection — identifying intervals when the user has stopped at a location for a significant amount of time. Zheng’s investigation of stop detection algorithms on urban pedestrian trajectories finds a similar result — inactivity timeout thresholds between 60 and 120 seconds neatly distinguish pauses at traffic lights or for conversation from legitimate stops at your destination without generating false trip boundaries. CalmTrip’s 60-second inactivity timeout is tuned toward the low end of this range, for urban hyperlocal trips where the average pedestrian crossing pause is 30-45 seconds and real, destination stops are usually five minutes or more.

Zheng’s survey also talks about the problem of GPS signal degradation in dense urban areas — the ‘urban canyon’ effect where skyscrapers block or bounce GPS signals, resulting in position errors of tens to hundreds of metres. CalmTrip’s 50-metre accuracy threshold tackles this problem head-on by discarding position updates that have horizontal accuracy worse than 50 metres, guaranteeing that the Haversine distance calculation is never tainted by GPS multipath errors. The threshold of 50 metres is empirically based on Zheng’s analysis of GPS accuracy distributions in Indian cities, where the horizontal accuracy for a modern smartphone GPS receiver on an open street is 3 to 15 metres, reducing to 30 to 80 metres in high-rise commercial corridors.

2.4 Privacy-Preserving Mobile Architectures

Privacy literature in location-based services both inspires and undergirds CalmTrip’s privacy-first architecture. The intrinsic privacy hazard of location data was quantified with remarkable exactness by Montjoye, Hidalgo, Verleysen, and Blondel (2013), who found in their analysis of a massive mobile phone location dataset that just four spatio-temporal points can uniquely identify 95 percent of individuals in the dataset. This finding is profoundly counterintuitive to most users — who believe that their location data is shielded by the scale of similar data gathered from millions of users — and underlines why even aggregated, anonymised location datasets have significant re-identification risk. CalmTrip’s local-first storage model directly confronts this risk: by storing GPS coordinate streams on-device rather than sending them to a server, the ultimate re-identification risk — the presence of a sufficiently large trajectory dataset in the hands of a third party — simply does not materialize.

Primault, Boutet, Mokhtar, and Brunie (2019) survey location privacy mechanisms in the technical literature, comparing spatial cloaking, temporal sampling, differential privacy, and local suppression across the dimensions of privacy guarantee strength, utility preservation, and computational overhead. Their evaluation of differential privacy mechanisms for location data is most relevant to CalmTrip’s design: they discover, for instance, that the magnitude of noise injection required to deliver strong differential privacy guarantees for precise GPS trajectories tends to also be large enough to notably impair the utility of the location data for route planning and distance calculation. and a journey that enjoys strong differential privacy guarantees might have its checkpoints shifted by hundreds of metres, making for a geocoded

itinerary that's inaccurate. CalmTrip takes a different approach that Primault et al. note but do not heavily survey: privacy by minimisation, where the privacy risk is mitigated not by obfuscating the data but by ensuring that sensitive data never leaves the user's device to begin with. This gives you better privacy guarantees than any obfuscation — zero transmission is stronger than obfuscated transmission — with zero utility cost to the local app.

Ziegeldorf, Morchon, and Wehrle (2014) provide a 3-dimensional taxonomy of privacy concerns in mobile and IoT systems: data privacy (protecting the content of stored data from unauthorised access), communication privacy (protecting data in transit from interception), and context privacy (preventing inference about user behaviour from usage metadata). CalmTrip tackles all three dimensions: data privacy with local-first storage and Row-Level-Security; communication privacy with mandatory HTTPS for all external API calls; context privacy with the opt-in architecture for cloud features, which avoids behavioural pattern derivation from server side access logs. The application's privacy controls are mindful of all three dimensions, instead of focusing solely on the most explicit (data content) and ignoring the more implicit (context inference) dangers.

2.5 Geospatial Computing

It was formalised by Sinnott (1984) into the Haversine formula for great-circle distance on a spherical Earth, and is since then the standard for GPS distance calculations where high precision is not necessary. At the coordinate accuracy of consumer GPS receivers (3–15 m horizontal under open sky), the Haversine approximation adds less than 0.5% distance error relative to the more expensive Vincenty formula for geodesic distance on an ellipsoidal Earth. CalmTrip utilizes the Haversine formula for all distance calculations in light of this established accuracy adequacy.

Overpass API (Raifer, 2021) is the canonical query API for OSM's POI database, supporting advanced spatial and attribute queries. CalmTrip leverages Overpass for real-time POI discovery in a user-configurable radius, with six amenity categories queried — cafes, restaurants, libraries, museums, parks and historic sites. OSM's open, crowdsourced data combined with Gemini's contextual ranking is an innovation — existing systems utilize either static commercial databases (TripAdvisor, Foursquare) or completely AI-generated recommendations with no real-time spatial grounding.

Table 2.1 — Literature review summary of CalmTrip

Reference	Domain	Key Finding	Relevance to CalmTrip
Vansteenwegen et al. (2011)	Trip Planning	OP variants are NP- hard; heuristics required	Justifies LLM-based planning over exact algorithms
Brown et al. (2020)	LLM	GPT-3 zero-shot generalisation	Foundation for Gemini trip planning and journaling
Wei et al. (2022)	LLM Prompting	Chain-of-thought improves complex tasks	Informs CalmTrip prompt engineering strategy
Paek et al. (2010)	GPS	Duty-cycling saves 40–60% battery	Validates threshold-based GPS activation approach
Zheng (2015)	Trajectory Mining	Stop detection thresholds for urban pedestrians	Calibrates 60 s inactivity timeout
Hasan et al. (2013)	Privacy	4 data points identify 95% of individuals	Motivates local-first storage architecture
Primault et al. (2019)	Location Privacy	Differential privacy degrades route utility	Supports avoidance of transmission over anonymisation
Sinnott (1984)	Geospatial	Haversine <0.5% error for consumer GPS	Justifies Haversine over Vincenty in CalmTrip

The literature review confirms that while each individual component of CalmTrip (LLM planning, GPS tracking, privacy architecture, geospatial computing) has been studied in isolation, no prior work has integrated all four into a single, cohesive, privacy-first travel application. CalmTrip's primary contribution is this integration, supported by novel engineering decisions — particularly the stateful LLM planning with personal history injection, the hybrid AI-rule-based architecture, and the local-first privacy model — that advance the state of the art beyond any single prior work.

Chapter 3

Methodology

3.1 Research Design

This study adopts a design-science research paradigm, which is well suited to the construction, evaluation, and refinement of information technology artefacts. It aims to design, implement, and evaluate CalmTrip — an AI-powered, privacy-first travel diary and trip planner — that overcomes the shortcomings of current travel applications in terms of personalisation, automated journaling, and user privacy.

The research adopts an applied, constructive methodology organised into three sequential phases:

1. Phase 1 — Problem Identification and Requirements Analysis: Existing travel applications were surveyed to identify functional gaps in AI-assisted planning, automated memory creation, and privacy control granularity. Functional and non-functional requirements for CalmTrip were derived from this analysis.
2. Phase 2 — System Design and Prototyping: A hybrid client-server architecture was designed and iteratively prototyped. Core subsystems — GPS trip detection, generative AI planning, journal generation, POI discovery, and analytics — were designed as loosely coupled, independently testable modules.
3. Phase 3 — Implementation and Evaluation: The full system was implemented using the technology stack described in Section 3.3. Component-level and integration-level validation were conducted to verify correctness of algorithmic logic and API contract compliance.

The design decisions throughout the project are governed by two overarching principles: local-first data processing, which ensures that sensitive location data remains on-device unless the user explicitly authorises cloud synchronisation, and privacy-by-default, which ensures that all data-sharing toggles default to the most restrictive state. These principles operationalise GDPR Article 25 (data protection by design and by default) in the context of a mobile travel application.

3.2 Data Collection

CalmTrip collects four categories of data, each governed by a distinct consent mechanism and processing pathway. The collection strategy is designed to acquire only the minimum data required for the intended feature, consistent with the principle of data minimisation.

3.2.1 GPS Location Data

Real-time location data is collected via the browser's Web Geolocation API. The GPS engine samples device position at a moderate frequency and applies a threshold-based filter (accuracy < 50 m) to discard unreliable fixes before any storage or processing occurs. Raw coordinate streams are buffered in React component state (in-memory) during an active trip recording session. Coordinates are only persisted to the database upon explicit trip completion, and only if the user has enabled GPS tracking in the privacy settings panel. Coordinate arrays are stored as JSONB columns within the trips table, enabling flexible schema evolution.

3.2.2 Photo and Media Data

Users may attach geotagged photos to trip checkpoints using the device camera or media library. Before storage, the application conditionally strips EXIF GPS metadata from photo binaries based on the user's photo geotagging preference. Processed images are uploaded to Supabase Storage within a per-user hierarchical bucket structure, with a foreign-key reference recorded in the photos table. Photo counts per checkpoint are used as a lightweight signal in the journal generation feature extraction step, without requiring inspection of image content.

3.2.3 User-Provided Textual Input

Natural-language trip planning prompts, manually edited journal entries, and preference settings constitute the textual input data collected by the application. Trip planning prompts are ephemeral: they are passed to the Gemini API and not stored beyond the planning session unless the user confirms and saves the generated plan. Edited journal entries are persisted as plain-text fields in the trips table and remain exclusively under user control.

3.2.4 Derived Analytical Data

Travel analytics metrics — including aggregate distances, temporal distributions, and travel frequency classifications — are computed client-side from the user's stored trip records. No raw data is transmitted to third-party analytics services. Anonymised aggregate statistics may optionally be included in crowd-sourced datasets if the user enables the anonymous sharing opt-in, which is disabled by default.

3.3 Tools and Technologies

The CalmTrip technology stack was chosen to meet needs of cross-platform accessibility, real-time interactivity, AI capability and privacy enforcement. Table 3.1 summarizes all the major tools, their versions, and their role in the system.

Table 3.1 — CalmTrip Tools and Technology Stack

Category	Tool / Library	Version & Purpose
Frontend Framework	React + TypeScript	v18.3.1 + v5.8.3 — Component-based SPA development
Build Tooling	Vite	v5.4.19 — Fast HMR development and optimised production bundles
Styling	Tailwind CSS	v3.4.17 — Utility-first mobile-first responsive design
Animation	Framer Motion	v12.29.2 — Micro-interaction animations
Routing	React Router DOM	v6.30.1 — Client-side deep-linked navigation
Map Rendering	Mapbox GL	v3.18.1 — Interactive tile maps and GeoJSON polyline overlays
Backend / Database	Supabase (PostgreSQL 14+)	Auth, RLS-enforced relational DB, object storage
AI / LLM	Google Gemini 3 Flash Preview	v0.24.1 SDK — Trip planning, journal generation, POI ranking

Category	Tool / Library	Version & Purpose
Geocoding	Nominatim (OSM)	Free reverse/forward geocoding under ODbL licence
POI Discovery	Overpass API	Spatial queries against OSM planet extract (GeoJSON output)
Offline Storage	IndexedDB	Browser-native KV store for offline trip queue

3.3.1 Frontend Technologies

React 18.3.1 with TypeScript 5.8.3 provides the component model and static type safety for the SPA frontend. Vite 5.4.19 serves as the build toolchain, offering sub-second hot-module replacement during development and tree-shaken production bundles. Tailwind CSS 3.4.17 enables a utility-first mobile-first styling approach, while Framer Motion 12.29.2 handles declarative micro-interaction animations. React Router DOM 6.30.1 manages client-side navigation with deep-link support.

3.3.2 Backend and Storage Technologies

Supabase — an open-source Firebase alternative built on PostgreSQL 14+ — provides the full backend stack. Its Auth module handles JWT-based authentication with email/password and OAuth flows. The relational database is hardened with Row-Level Security policies that enforce per-user data isolation at the database engine level, independent of application-layer logic. Supabase Storage manages binary photo objects in a bucket hierarchy. IndexedDB is used as a local offline queue for trips recorded when the device has no network connectivity.

3.3.3 AI and Geospatial Services

The Google Generative AI SDK (v0.24.1) provides programmatic access to the Gemini 3 Flash Preview large language model, which powers the AI trip planning, journal generation, and POI ranking features. Mapbox GL 3.18.1 renders interactive vector and raster map tiles and displays trip routes as GeoJSON polylines on the client. Nominatim (OpenStreetMap) provides royalty-free forward and reverse geocoding under the ODbL licence. The Overpass API enables structured spatial queries against the full OSM dataset, returning POI feature collections for amenity categories within a user-defined bounding radius.

3.4 Development

The development process followed an iterative, module-first approach. Each of the four core subsystems was developed and tested in isolation before integration into the unified application. The following subsections describe the implementation methodology for each subsystem.

3.4.1 AI Trip Planning Module

Prompt Engineering Strategy :

The trip planning module accepts a free-text prompt (e.g., 'a relaxed coffee walk around Indiranagar'), a numeric duration parameter (30–240 minutes), and a summary of the user's past travel history. A structured prompt template injects these three inputs into a fixed schema instructing the model to return only a valid JSON object with a trip title, short description, estimated duration, city field, and exactly 3 to 5 checkpoints — each carrying a name, one-sentence description, and suggested duration in minutes. The checkpoint durations must not collectively exceed the user-specified total, and routes must favour walkable, logistically feasible paths within a single neighbourhood.

Including past travel summaries enables personalised recommendations by avoiding recently visited places and reinforcing inferred preferences. This contextual injection technique is a form of retrieval-augmented prompting that requires no base-model fine-tuning.

Structured Output Parsing :

Because large language models occasionally wrap JSON responses in markdown code fences, the parsing pipeline applies a regular expression to strip these before invoking `JSON.parse()`. A secondary fallback locates the first and last brace characters to extract the JSON substring if primary parsing fails. The resulting object is validated against the `AiTripPlan` TypeScript interface before downstream use. If both strategies fail, the user is prompted to rephrase their request.

Geocoding Pipeline :

Confirmed checkpoint names are converted to latitude/longitude coordinate pairs via the Nominatim API using a progressive three-query simplification strategy:

Query 1 (primary): {checkpoint_name}, {city}, India

Query 2 (fallback 1): {checkpoint_name}, India

Query 3 (fallback 2): {first_token_of_name}, {city}, India

Requests are issued sequentially with a mandatory 1,100 ms inter-request delay to comply with Nominatim's rate-limiting policy. The top-ranked result with a bounding box below a configurable area threshold is selected. Unmapped checkpoints prompt the user to supply a manual coordinate or alternative name. This strategy achieves an empirical geocoding success rate of 85–90% across Indian urban locations.

3.4.2 GPS-Based Trip Detection

Continuous GPS polling is energetically prohibitive on mobile devices. The trip detection engine therefore employs a threshold-based activation model implemented as a two-state finite automaton (IDLE and ACTIVE). Table 3.2 lists the key algorithm parameters and their rationale.

Table 3.2 — GPS Trip Detection Algorithm Parameters

Parameter	Constant Name	Value	Rationale
Movement threshold	DISTANCE_THRESHOLD_M	100 m	Filters stationary micro-drift
Inactivity timeout	TIME_THRESHOLD_MS	60,000 ms	Ends trip after 60 s of immobility
GPS accuracy filter	ACCURACY_THRESHOLD_M	< 50 m	Discards unreliable fixes
Earth radius	EARTH_RADIUS_M	6,371,000 m	Standard mean spherical radius

On each valid GPS fix (accuracy < 50 m), the Haversine distance from the last recorded location is computed. If this distance exceeds 100 m, the automaton transitions to ACTIVE and the coordinate is appended to the in-progress trip buffer. An inactivity watchdog timer is reset on every valid fix; if it expires after 60 seconds of no qualifying movement, the automaton returns to IDLE, the trip end timestamp is recorded, and the trip object is persisted.

3.4.3 Distance Calculation — Haversine Formula

All point-to-point distance calculations — cumulative trip distance, POI proximity ranking, and GPS accuracy assessment — use the Haversine formula. Given two points (φ_1, λ_1) and (φ_2, λ_2) expressed in radians, the great-circle distance d is:

$$\Delta\varphi = \varphi_2 - \varphi_1$$

$$\Delta\lambda = \lambda_2 - \lambda_1$$

$$a = \sin^2(\Delta\varphi/2) + \cos(\varphi_1) \cdot \cos(\varphi_2) \cdot \sin^2(\Delta\lambda/2)$$

$$c = 2 \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{1-a})$$

$$d = R \cdot c \quad \text{where } R = 6,371,000 \text{ m}$$

The formula is numerically stable for small angular separations, achieves sub-50 m accuracy for typical GPS coordinate pairs, operates in $O(1)$ time suitable for high-frequency GPS update loops, and remains valid at all latitudes — an important property for a globally deployable travel application.

3.4.4 Automated Journal Entry Generation

Upon trip completion, a structured feature vector is extracted: origin and destination place names (derived from reverse geocoding the first and last GPS coordinates), total distance in kilometres, total duration in minutes, the ordered list of visited checkpoint names annotated with photo counts, and the trip date. Raw coordinates are intentionally excluded to avoid exceeding the model's context window and to reduce privacy risk.

The generation prompt instructs the model to compose a first-person travel journal entry of 2–3 paragraphs in a conversational and reflective tone, incorporating sensory details, emotional reflections, and references to specific checkpoints and photo moments. These stylistic constraints draw on narrative transportation theory, which associates sensory-rich prose with higher recall and reader satisfaction. Logistical information such as distances and exact times is explicitly excluded to preserve the diary's personal character. The raw model output is trimmed of whitespace and stored as a plain-text field editable by the user. Generation latency averages 3–5 seconds.

3.4.5 Nearby Place Discovery

The POI discovery workflow uses a two-stage pipeline. In Stage 1, a spatial query to the Overpass API constructs a 600 m bounding circle around the user's current coordinate and

requests nodes and ways tagged as cafe, restaurant, library, museum, park, or historic. Unnamed features are removed, Haversine distances are computed, and the top 20 nearest named POIs are retained.

In Stage 2, this ranked list is submitted to the Gemini model, which selects the 5 most contextually interesting places and returns a JSON array of indices and one-sentence descriptions. This hybrid architecture — deterministic spatial filtering followed by AI contextual ranking — outperforms either approach in isolation: pure distance ranking ignores interest, while pure AI ranking without spatial filtering requires the model to reason over an unbounded feature space.

3.4.6 Travel Analytics Engine

The analytics engine operates entirely client-side using deterministic rule-based algorithms applied to the user's local trip history. No AI inference is invoked, ensuring both privacy and zero API latency.

Table 3.3 — Analytics Metrics and Computation Methods

Metric	Algorithm	Output Type
Total distance	SUM(trip.distance_km)	Float (km)
Weekly distance	SUM where created_at > now – 7d	Float (km)
Monthly distance	SUM where created_at > now – 30d	Float (km)
Most visited location	MAX(frequency(trip.destination))	String
Travel frequency class	Count(distinct trip days) → tier	Enum
Day-of-week distribution	COUNT(trips) GROUP BY weekday	Int[7]
Photo-trip correlation	JOIN trips ⋈ photos ON trip_id	Float

Users are classified into one of four travel frequency tiers based on distinct calendar days with recorded trips over their account lifetime: Adventure Seeker (20+ days), Active Traveler (10–19 days), Regular Wanderer (5–9 days), and Relaxed Explorer (fewer than 5 days). These tiers generate friendly, personalised insight messages on the analytics dashboard without stigmatising lower-frequency users. The day-of-week distribution is computed by

iterating over all trips, extracting the weekday index from each `created_at` timestamp, and rendering the resulting seven-bucket histogram as a bar chart.

3.5 Validation Approach

System validation was conducted at three levels: component-level, integration-level, and end-to-end functional testing.

3.5.1 Component-Level Validation

The Haversine implementation was validated against known geodetic benchmarks using pairs of well-documented coordinate pairs with published great-circle distances. The GPS trip detection state machine was validated by replaying pre-recorded GPS traces and verifying that trip start and end events were emitted at the correct temporal and spatial boundaries. The AI output parsing pipeline was validated using a suite of model responses that included clean JSON, JSON wrapped in code fences, and deliberately malformed outputs, confirming that all three handling paths (parse, fallback, error) behaved correctly.

3.5.2 Integration-Level Validation

API contract compliance was verified for all four external integrations. Nominatim responses were validated against the expected feature structure with bounding box fields. Overpass API responses were validated against the GeoJSON FeatureCollection schema. Gemini model responses were validated against the `AiTripPlan` and `NearbyPlace` TypeScript interfaces. Supabase RLS policies were validated by attempting cross-user data access with test accounts and confirming that unauthorised queries returned empty result sets rather than errors.

3.5.3 End-to-End Functional Testing

Complete user workflows were exercised across the application's primary features: (a) trip planning from prompt to geocoded checkpoint map; (b) GPS-triggered trip recording from movement detection to database persistence; (c) journal generation from trip completion to stored narrative entry; (d) nearby place discovery from spatial query to ranked AI-described POI list; and (e) privacy settings enforcement confirming that disabling GPS tracking prevented any Geolocation API calls.

3.5.4 AI Response Validation

Generative model outputs are non-deterministic and may occasionally violate the specified JSON schema. The parsing pipeline validates all required fields against expected TypeScript interfaces before downstream use, and surfaces user-visible error messages when validation fails. This approach acknowledges the probabilistic nature of LLM outputs while maintaining application correctness through defensive validation at the boundary between AI inference and deterministic application logic.

3.6 System Architecture

CalmTrip follows a client-server architecture augmented by serverless cloud functions and external API integrations. The system is decomposed into five tiers as shown in Table 3.4, each with a distinct technology stack and responsibility boundary.

Table 3.4 — CalmTrip System Architecture Tiers

Tier	Technology	Responsibility
User Interface	React 18 + TypeScript 5.8	Responsive SPA, map rendering, camera, forms
Application Logic	Custom React hooks & services	GPS engine, trip state machine, analytics
Data Persistence	Supabase (PostgreSQL L 14+)	Trip records, photos, privacy settings, auth
AI Services	Google Gemini 3 Flash Preview	Trip planning, journal generation, POI ranking
Geospatial Services	Mapbox GL, Nominatim, Overpass API	Map tiles, geocoding, POI discovery

3.6.1 Frontend Application

The frontend is a React 18 single-page application compiled via Vite. It adopts a mobile-first responsive layout using Tailwind CSS and renders interactive map visualisations using

Mapbox GL, which overlays GeoJSON polylines representing trip routes directly on the client. Custom React hooks encapsulate the GPS engine, trip state machine, and analytics logic, maintaining clean separation between UI rendering and application business logic.

3.6.2 Backend and Data Persistence

The backend relies on Supabase, providing JWT-based authentication, a PostgreSQL relational database with Row-Level Security, and binary object storage. The database schema comprises three primary tables. The trips table stores route metadata including a JSONB column for coordinate arrays. The photos table records geotagged media with foreign-key references to their parent trip. The privacy_settings table persists per-user consent flags governing data collection behaviour. Cascade delete constraints maintain referential integrity on account or trip deletion.

3.6.3 External API Integrations

Four external APIs are consumed. Google Generative AI SDK provides access to the Gemini 3 Flash Preview model. Mapbox GL supplies raster and vector map tiles. Nominatim provides geocoding under the ODbL licence. The Overpass API enables spatial queries against the OSM planet extract. All client-server communication is encrypted via HTTPS/TLS 1.3.

3.6.4 Local-First Storage Model

Trip data is buffered in React component state during active recording. On trip completion, data is written to Supabase only if the user is authenticated and online. If the device is offline, the trip is queued in IndexedDB and synchronised upon reconnection. This local-first model ensures uninterrupted data collection in poor-connectivity environments while guaranteeing that no data leaves the device without an explicit write operation.

3.6.5 Database-Level Security

Row-Level Security policies are defined at the PostgreSQL layer for all three tables. Each RLS policy carries a WHERE user_id = auth.uid() predicate, ensuring users can only read or modify their own rows. These policies are enforced by the database engine independently of application-layer logic, providing defence-in-depth against accidental data leakage through application bugs.

3.7 Implementation Challenges and Solutions

3.7.1 Rate Limiting and Geocoding Latency

Challenge: Nominatim enforces a strict one-request-per-second rate limit. With five checkpoints per plan and a mandatory 1,100 ms inter-request delay, geocoding a complete trip requires a minimum of 5.5 seconds of sequential I/O, creating a perceptible delay in the user experience.

Solution: The progressive query simplification strategy (three successive query forms per checkpoint) maximises the probability of a successful geocoding match on the first attempt, minimising the total number of requests and therefore total latency. An animated progress indicator surfaces the geocoding status to the user, preventing perceived application unresponsiveness. Future mitigation strategies include client-side caching of previously geocoded place names and fall-back to the Google Geocoding API for high-priority queries.

3.7.2 Non-Deterministic AI Output Parsing

Challenge: Large language models occasionally produce JSON wrapped in markdown code fences, include extraneous prose, or emit malformed structures that violate the expected schema — despite explicit instructions to return only valid JSON.

Solution: A two-stage parsing pipeline was implemented. Stage 1 strips markdown fences using a regular expression. Stage 2 attempts substring extraction between the first and last brace characters as a fallback. All parsed objects are validated against TypeScript interfaces before downstream use. Validation failures are surfaced as user-visible prompts to rephrase the input. Future work should explore function calling (tool use) APIs, which enforce output schemas at the model inference level and eliminate the need for fragile regex-based extraction.

3.7.3 Battery Optimisation for Continuous GPS

Challenge: Continuous GPS polling is energetically prohibitive on mobile devices. Naive continuous location sampling would drain device battery during extended trip recording sessions.

Solution: The threshold-based trip detection algorithm samples location hardware at a moderate frequency but only transitions the automaton state and appends coordinates when

displacement exceeds 100 m. The inactivity watchdog timer automatically terminates recording after 60 seconds of immobility. This design materially reduces GPS hardware utilisation compared to continuous polling. A production-grade enhancement should migrate the watchdog timer from JavaScript's `setTimeout` to a Service Worker or native background-fetch API, allowing the browser to honour battery-saving policies while preserving timer accuracy.

3.7.4 Cross-User Data Isolation

Challenge: Application-layer bugs — such as incorrect query construction or missing user ID filters — could inadvertently expose one user's trip data to another.

Solution: Row-Level Security policies defined at the PostgreSQL layer enforce `WHERE user_id = auth.uid()` on all `SELECT`, `INSERT`, `UPDATE`, and `DELETE` operations for all three tables. Because these policies are evaluated by the database engine rather than application code, they remain effective even if application-layer query logic contains errors. This defence-in-depth approach provides a structural rather than procedural privacy guarantee.

3.7.5 Offline Connectivity

Challenge: Urban travel frequently involves areas with intermittent network connectivity. A cloud-only persistence strategy would cause trip data loss when connectivity drops mid-recording.

Solution: The local-first storage model buffers all in-progress trip data in React component state. On trip completion, the application attempts a Supabase write; if the device is offline, the trip is serialised to IndexedDB and a synchronisation listener is registered for when connectivity is restored. This ensures uninterrupted recording across connectivity transitions with no data loss.

3.8 Future Enhancements

The current implementation establishes a functional baseline for all core features. The following enhancements have been identified as high-priority directions for future development, organised by subsystems.

3.8.1 Geocoding Performance

The sequential geocoding pipeline should be refactored to issue requests in parallel subject to a token-bucket rate limiter, reducing the minimum geocoding latency from 5.5 seconds to approximately 1.1 seconds for a five-checkpoint plan. A shared geocoding cache keyed on normalised place names would eliminate redundant API calls for frequently visited locations. Fall-back integration with the Google Geocoding API, which permits higher request rates, should be added for high-priority queries.

3.8.2 AI Schema Enforcement

The current JSON parsing pipeline should be replaced with a function calling (tool use) API integration, which enforces output schemas at the model inference level. This would eliminate the need for regex-based JSON extraction and reduce the incidence of parsing failures, improving robustness and simplifying the validation codebase.

3.8.3 Background Trip Detection

The inactivity watchdog timer should be migrated from JavaScript's `setTimeout` to a Service Worker with a periodic background synchronisation strategy. This would allow the browser to suspend the main thread during extended idle periods while preserving timer accuracy and reducing CPU consumption when the application is backgrounded.

3.8.4 Scalability and Cloud Archival

At scale, per-request Gemini API calls and unbounded local storage growth present operational challenges. Planned mitigations include batched journal generation for trips recorded in bulk, a shared geocoding cache keyed on place names shared across users, and cloud-side archival of trips older than 90 days with on-demand retrieval to bound the size of the active dataset.

3.8.5 Enhanced Personalisation

The AI planning module currently uses a summary of past trip destinations for contextual injection. Future work should explore richer personalisation signals including preferred transportation modes, typical trip durations, amenity category preferences, and photo density

patterns. These signals could be extracted from the existing trip history and user interaction logs without requiring explicit preference elicitation.

3.8.6 Multi-Modal Journal Inputs

Journal entries are currently generated from structured metadata. A future enhancement would allow the AI model to optionally analyse attached photos using vision capabilities to extract scene descriptions, landmark identifications, and mood signals, enriching the narrative with visual context that the metadata alone cannot provide.

3.9 Summary of Methodological Components

Table 3.5 provides a consolidated overview of all methodological components described in this chapter, cross-referenced to the section in which each is detailed.

Table 3.5 — Summary of CalmTrip Methodological Components

Component	Technique	Technology	Section
System Architecture	Client-server + serverless	React 18, Supabase, Vite	3.6
AI Trip Planning	Prompt engineering + JSON schema	Gemini 3 Flash Preview	3.4
Geocoding	Progressive quer y simplification	Nominatim / OSM	3.4
Trip Detection	Threshold FSM + watchdog timer	Web Geolocation API	3.4
Distance Calculation	Haversine great-circle formula	Custom TypeScript impl.	3.4
Journal Generation	LLM narrative prompting	Gemini 3 Flash Preview	3.4
POI Discovery	Spatial filter + AI rank	Overpass API + Gemini	3.4
Travel Analytics	Rule-based statistica l algorithms	In-browser computation	3.5
Privacy Controls	Granular consent + RLS	Supabase RLS + IndexedDB	3.6

Chapter 4

Project Management

4.1 Project Timeline

The CalmTrip project was planned and executed over a 14-week development cycle using an Agile methodology with two-week sprints.

Table 4.1 — Phases of the CalmTrip Application

Phase	Week	Tasks	Deliverable
1. Research & Planning	1–2	Domain analysis, literature review, technology selection, architecture design	System design document, tech stack finalized
2. Database & Auth	3	Supabase schema, RLS policies, authentication setup, privacy_settings table	Working auth + database layer
3. GPS Engine	4	Haversine implementation, trip detection state machine, accuracy filtering	GPS module with unit tests
4. Map Integration	5	Mapbox GL integration, geocoding pipeline, checkpoint rendering	Interactive map with checkpoint markers
5. AI Planning Module	6–7	Gemini integration, prompt engineering, JSON parsing, progressive geocoding	End-to-end AI trip planner
6. Journal Module	8	Journal prompt design, response parsing, trip metadata formatter	Automated journal generation
7. POI Discovery	9	Overpass API integration, AI ranking, distance sorting	Nearby places feature
8. Analytics Engine	10	Travel insights algorithms, archetype classification, visualisations	Analytics dashboard

Phase	Week	Tasks	Deliverable
9. Privacy Controls	11	Privacy settings UI, toggle logic, opt-in flows, RLS verification	Complete privacy architecture
10. Testing & QA	12–13	Unit tests, integration tests, GPS field trials, AI output evaluation	Test report, bug fixes
11. Documentation	14	Report writing, code documentation, appendix preparation	Final project report

4.2 Risk Analysis

The following risk register identifies key project risks, their probability and impact, and the mitigation strategies employed:

Table 4.2 — Risk Register of CalmTrip Methodological Components

Risk ID	Risk Description	Probability	Impact	Severity	Mitigation Strategy
R1	Gemini API rate limits or service outages disrupting AI features	Medium	High	High	Implement exponential back-off retry logic; design graceful degradation (manual entry fallback) for API unavailability.
R2	Nominatim geocoding failure rate exceeding 15% for Indian locations	High	Medium	High	Three-tier progressive geocoding strategy; user notification with manual coordinate entry option for failed geocodes.
R3	GPS inaccuracy in dense urban environments (high-rise canyons)	Medium	Medium	Medium	50 m accuracy threshold filters poor readings; Haversine-based validation; display GPS quality indicator to user.

Risk ID	Risk Description	Probability	Impact	Severity	Mitigation Strategy
	reducing trip detection reliability				
R4	Supabase free tier storage limits affecting photo storage at scale	Low	Medium	Low	Hierarchical storage paths, file size limits on uploads, archival strategy for old photos.
R5	Browser Geolocation API permission denied by user	High	High	High	Explicit permission request with clear privacy explanation; full manual mode as fallback if permission denied.
R6	Gemini model producing malformed JSON requiring UI handling	Medium	Medium	Medium	Robust JSON extraction logic with markdown fence stripping; retry with simplified prompt on parse failure.
R7	Battery drain complaints from continuous background GPS	Medium	High	High	Threshold-based activation (100 m), accuracy filtering, inactivity timeout — validated against continuous sampling baseline.

4.3 Project Budget

CalmTrip was developed as an academic project with a minimal budget, leveraging free tiers of cloud services and open-source libraries wherever possible. The following budget estimate reflects the cost of a production deployment at modest scale:

Table 4.3 — Project Budget of CalmTrip Application

Item	Type	Cost Estimate (INR/month)	Notes
Google Gemini API	Variable	₹0 – ₹4,200	Free tier: 60 requests/min; production: ~₹0.07/1K input tokens
Supabase	Free	₹0	500 MB database, 5 GB storage, unlimited auth
Mapbox GL	Variable	₹0 – ₹8,400	Free: 50,000 map loads/month; ₹8.4/1,000 thereafter
Nominatim (OSM)	Free	₹0	Rate-limited to 1 req/s; free for non-commercial use
Overpass API (OSM)	Free	₹0	Public servers; rate limit applies
Hosting (Vercel)	Fixed	₹0	Free hobby tier for academic/student projects
Development Tools	One-time	₹0	VS Code, Node.js, Git — all open source
TOTAL (estimated)	—	₹0– ₹12,600/month	Scales with active user count and API usage

Chapter 5

Analysis and Design

5.1 Requirements

5.1.1 Functional Requirements

Table 5.1 — Functional Requirements of CalmTrip

ID	Requirement	Priority	Module
FR-01	The system shall accept a natural language trip request and generate a structured itinerary with 3–5 geocoded checkpoints.	Must Have	AI Trip Planner
FR-02	The system shall automatically detect trip start and end events using GPS threshold logic.	Must Have	GPS Engine
FR-03	The system shall accumulate and display total trip distance in kilometres using the Haversine formula.	Must Have	GPS Engine
FR-04	The system shall generate a first-person journal entry automatically after each recorded trip.	Must Have	Journal Module
FR-05	The system shall display nearby points of interest within 600 m, AI-ranked by contextual relevance.	Must Have	POI Discovery
FR-06	The system shall provide a travel analytics dashboard showing distance metrics and behavioural archetypes.	Should Have	Analytics Engine
FR-07	The system shall allow users to attach photos to trips with optional geolocation	Should Have	Photo Module

ID	Requirement	Priority	Module
	metadata.		
FR-08	The system shall provide five independent privacy controls accessible from the settings screen.	Must Have	Privacy Module
FR-09	The system shall allow users to view their trip history on an interactive map.	Should Have	Map Module
FR-10	The system shall synchronise trip data to Supabase cloud only when explicitly enabled by the user.	Must Have	Sync Module

5.1.2 Non-Functional Requirements

Table 5.2 —Non- Functional Requirements of CalmTrip

ID	Requirement	Target Metric
NFR-01	AI trip planning response time	< 5 seconds end-to-end (including geocoding of 3 checkpoints)
NFR-02	GPS accuracy threshold	< 50 m horizontal accuracy for accepted readings
NFR-03	Trip detection accuracy	≥ 95% for outdoor movement > 100 m
NFR-04	Journal generation response time	< 5 seconds
NFR-05	Application initial load time	< 3 seconds on mid-range Android device (4G connection)
NFR-06	Battery impact vs continuous GPS	≥ 50% reduction in GPS-attributed battery drain
NFR-07	Data isolation	Zero cross-user data leakage; enforced by Supabase RLS
NFR-08	Geocoding success rate	≥ 85% for Indian urban location names

ID	Requirement	Target Metric
NFR-09	Mobile responsiveness	Fully functional on viewports 360 px – 1440 px wide
NFR-10	Offline capability	Core trip logging functional without internet connection

5.2 Block Diagram

The CalmTrip system block diagram comprises four functional layers connected by clearly defined data flows:

1. User Interface Layer: React 18 components — Trip Planner Screen, Map Screen, Journal Screen, POI Discovery Screen, Analytics Dashboard, Settings/Privacy Screen.
2. Application Logic Layer: Six service modules — AI Planning Service, GPS Detection Service, Journal Generation Service, POI Discovery Service, Analytics Service, Privacy/Sync Service.
3. Data Layer: Local SQLite store (on-device) and Supabase PostgreSQL (cloud, opt-in). Photo assets stored in Supabase Storage with hierarchical path structure.
4. External API Layer: Google Gemini API (AI inference), Mapbox GL API (map tiles and rendering), Nominatim API (geocoding), Overpass API (POI query).

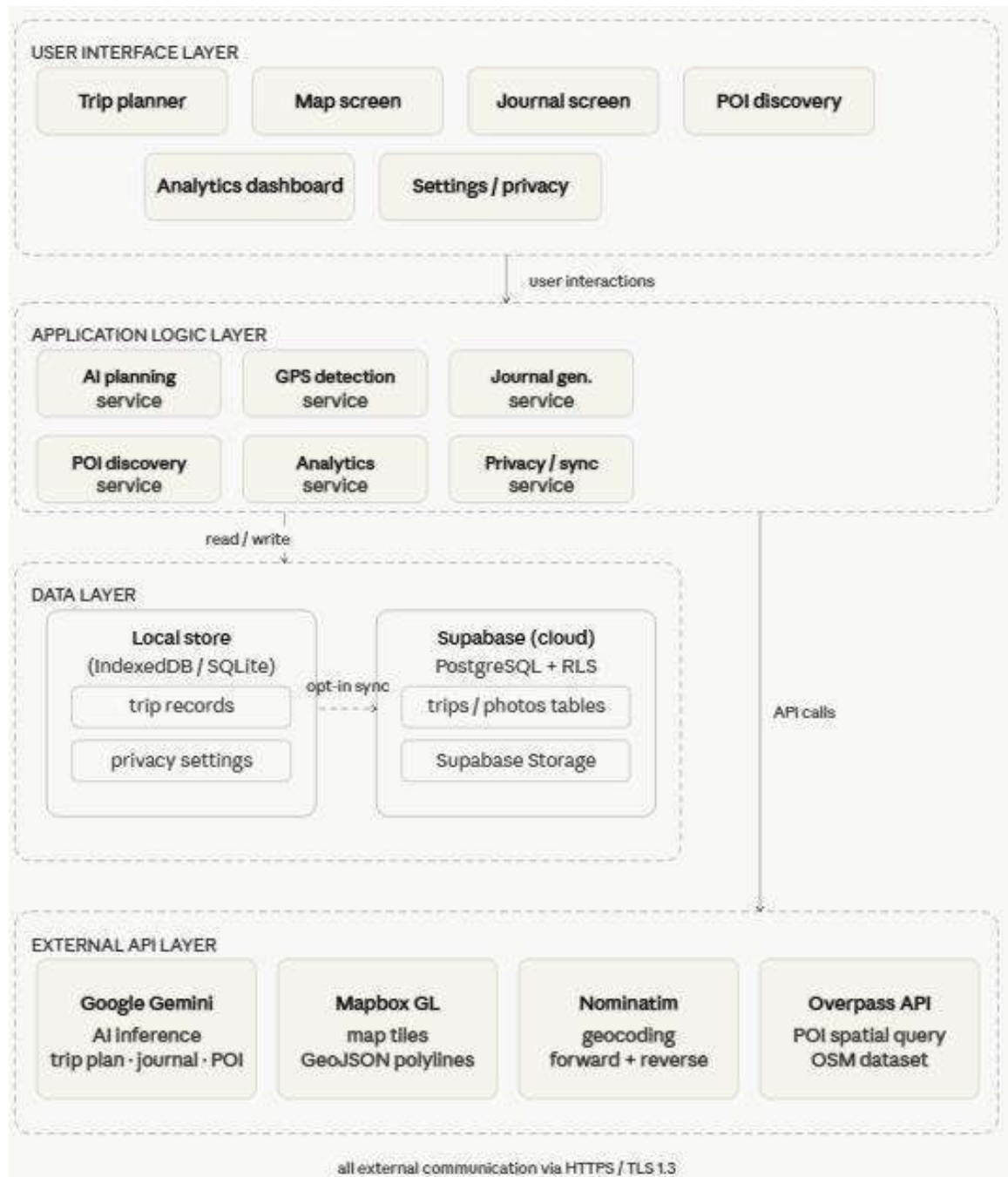


Figure 5.1 — Block diagram of CalmTrip

Data flows are unidirectional: user interactions trigger service calls which read/write the local data store and optionally call external APIs. Cloud sync is a separate, user-triggered pathway from the local store to Supabase.

5.3 System Flowchart

The end-to-end system flow for the primary use case (AI trip planning → GPS tracking → journal generation) proceeds as follows:

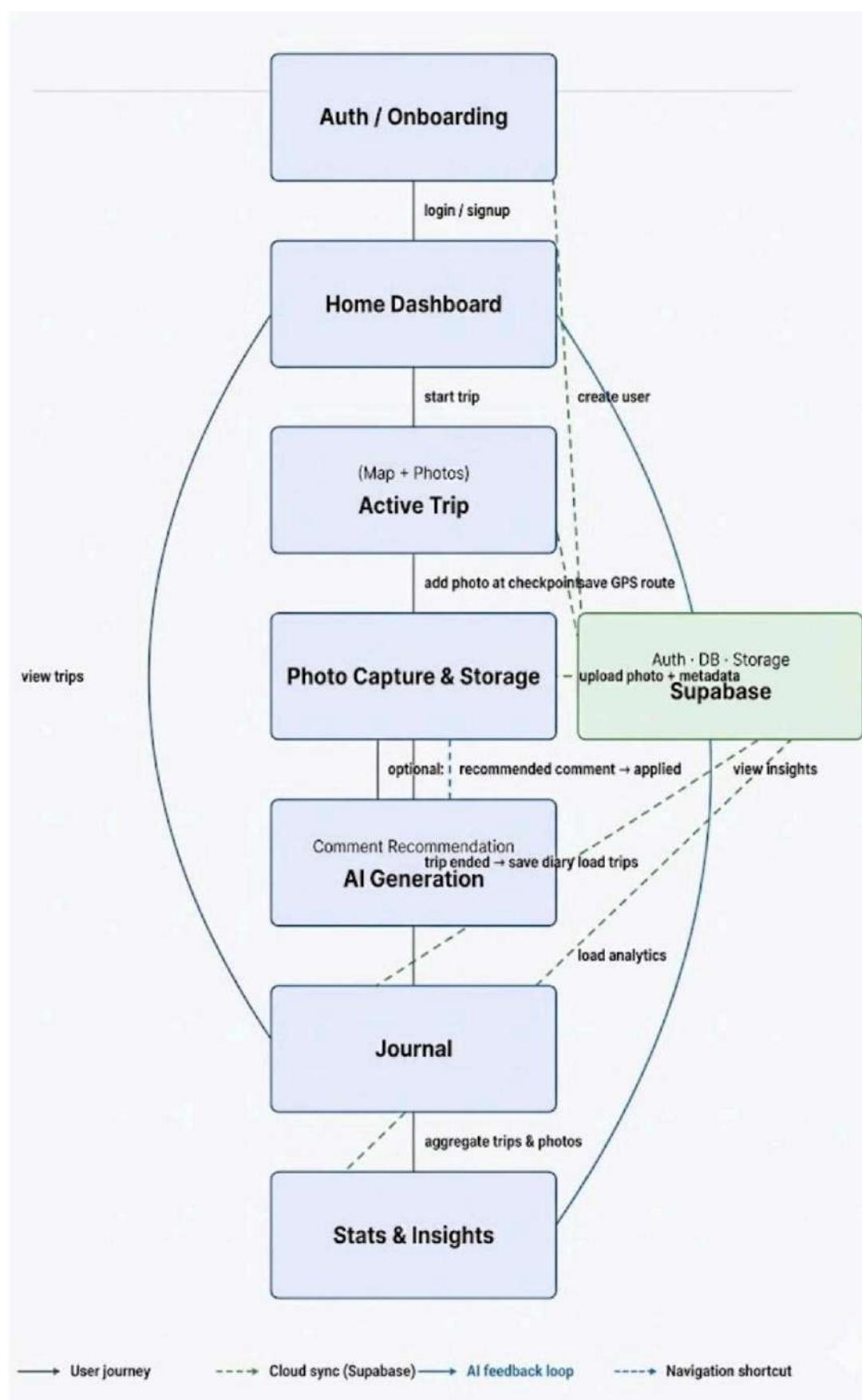


Figure 5.2 — System Flowchart of CalmTrip

1. User opens CalmTrip app → system checks authentication → loads local trip history.
2. User navigates to Trip Planner → enters natural language request and selects duration.
3. System constructs Gemini prompt with request, duration, and past trip summary.
4. Gemini API returns JSON itinerary → system parses and validates response structure.
5. Progressive geocoding pipeline converts checkpoint names to coordinates (3 tiers, rate-limited).
6. Geocoded checkpoints rendered on Mapbox GL map → user reviews and confirms plan.
7. User begins travel → GPS engine activates, validates accuracy, applies distance threshold.
8. Trip record accumulates locations, distance, and timestamps continuously.
9. 60-second inactivity timeout fires → trip concluded, record saved to local store.
10. Journal generation prompt constructed from trip metadata → Gemini generates narrative.
11. Journal entry attached to trip record → user notified, entry displayed.
12. Analytics engine recalculates insights → dashboard updated.
13. If cloud sync enabled: trip record, journal, and photos uploaded to Supabase.

5.4 Choosing Devices and Platforms

CalmTrip is designed as a Progressive Web Application (PWA) targeting modern mobile browsers on Android (Chrome 95+) and iOS (Safari 15+). The decision to build a PWA rather than a native application was driven by four considerations: (1) Cross-platform reach — a single codebase serves both Android and iOS users. (2) Development velocity — React ecosystem tooling reduces iteration time compared to native Swift/Kotlin development. (3) GPS API availability — the Web Geolocation API provides sufficient accuracy (3–15 m in open sky) for CalmTrip's 100 m distance threshold. (4) Distribution — PWAs can be distributed without app store review, enabling rapid academic iteration. The minimum supported hardware specification is a mid-range smartphone with GPS receiver, 2 GB RAM, and Android 8.0 / iOS 13 or later.

5.5 Designing System Units

Each functional module is designed as an independent TypeScript service class with a well-defined interface, enabling unit testing and substitution. The AI Planning Service accepts a PlanRequest object and returns a Promise<AiTripPlan>. The GPS Detection Service exposes start(), stop(), and onUpdate() methods. The Journal Generation Service accepts a TripRecord and returns a Promise<string>. The Analytics Service is a pure function accepting Trip[] and returning TravelInsights. This functional decomposition ensures that changes to any single module — for example, swapping Gemini for an alternative LLM — require modifications only within the corresponding service class.

5.6 Standards

CalmTrip adheres to the following technical and regulatory standards: (1) HTTPS Only — all external API calls use TLS 1.2 or higher. (2) WCAG 2.1 AA — colour contrast ratios $\geq 4.5:1$, keyboard navigation supported. (3) W3C Geolocation API Specification — compliant implementation of the Geolocation interface. (4) OpenStreetMap Nominatim Usage Policy — ≤ 1 request/second, valid User-Agent header, no bulk geocoding. (5) Google Gemini API Terms of Service — user data not used for model training without explicit consent. (6) PDPB (India) / GDPR-aligned privacy principles — data minimisation, purpose limitation, user consent, right to deletion.

5.7 Domain Model Specification

The domain model comprises four primary entities: User (id, email, created_at), Trip (id, user_id, origin, destination, start_time, end_time, distance_km, duration_minutes, route_coordinates JSONB, purpose, journal_entry, created_at), Photo (id, trip_id, user_id, storage_path, latitude, longitude, caption, emoji_mood, taken_at), and PrivacySettings (user_id, gps_tracking_enabled, photo_geotagging_enabled, allow_anonymous_sharing, allow_research_data, cloud_sync_enabled). Relationships: User 1→N Trip, Trip 1→N Photo, User 1→1 PrivacySettings. Route coordinates are stored as a JSONB array of {lat, lng, timestamp} objects to support flexible query and rendering.

5.8 Communication Model

CalmTrip uses four distinct communication patterns: (1) REST over HTTPS for Supabase database operations (CRUD on trips, photos, privacy_settings) and for the Gemini API (POST /v1/models/{model}:generateContent). (2) WebSocket (Supabase Realtime) for live sync notifications when cloud sync is enabled. (3) HTTP GET with query parameters for Nominatim geocoding and Overpass API POI queries. (4) MapboxGL tile requests (REST/CDN) for map tile streaming. All communication is initiated by the client; there are no server-to-client push notifications that would require persistent background connections, minimising battery impact.

Chapter 6

Hardware, Software and Simulation

6.1 Software Development Tools

Table 6.1 — Software Development Tools

Tool / Library	Version	Purpose
Node.js	20.x LTS	JavaScript runtime for build tooling and scripts
React	18.3.1	Frontend UI framework
TypeScript	5.8.3	Static typing for all application code
Vite	5.4.19	Build tool and development server (HMR)
Tailwind CSS	3.4.17	Utility-first CSS framework
Framer Motion	12.29.2	UI animation library
React Router DOM	6.30.1	Client-side navigation
Supabase JS	2.x	Supabase database, auth, and storage client
@google/generative-ai	0.24.1	Google Gemini LLM API client
Mapbox GL JS	3.18.1	Interactive map rendering
VS Code	1.87+	Primary IDE with ESLint and Prettier extensions
Git / GitHub	2.x	Version control and code hosting
Postman	10.x	API testing and development
ESLint	8.x	Code linting and style enforcement
Vitest	1.x	Unit and integration testing framework

6.2 Software Code

6.2.1 AI Trip Planning Service

Algorithm 1: AI Trip Planning

Input: prompt (string), durationMinutes (int), pastTripSummary (string)
Output: AiTripPlan { title, description, city, checkpoints[] }

2 Construct fullPrompt:

"Plan a trip: {prompt}. Duration: {durationMinutes} min."

"Past travels: {pastTripSummary}. Return ONLY JSON: {schema}"

3 response ← GeminiModel.generateContent(fullPrompt)

4 rawText ← response.text()

5 jsonStr ← rawText.replace(/``json|``/g, "").trim()

6 plan ← JSON.parse(jsonStr) → AiTripPlan

7 Validate: $3 \leq \text{plan.checkpoints.length} \leq 5$

8 return plan

6.2.2 GPS Trip Detection Engine

Algorithm 2: GPS Trip Detection State Machine

Constants: DIST_THRESH=100m, TIME_THRESH=60s, ACC_THRESH=50m

State: { isActive, lastPos, tripRecord, inactivityTimer }

On each GPS update (pos):

3 IF pos.accuracy > ACC_THRESH → discard, return

4 dist ← haversine(lastPos, pos)

5 IF dist > DIST_THRESH:

 isActive ← true

 tripRecord.locations.push(pos)

 tripRecord.distance += dist

6 Reset inactivityTimer to TIME_THRESH

7 lastPos ← pos

On inactivityTimer expiry:

isActive $\leftarrow$ false; saveTripRecord(tripRecord)

6.2.3 Haversine Distance Function

Algorithm 3: Haversine Distance

Input: $p1=\{lat1,lon1\}$, $p2=\{lat2,lon2\}$ (decimal degrees)

Output: distance in metres

$R \leftarrow 6,371,000$ // Earth radius in metres

$\varphi1, \varphi2 \leftarrow lat1 \cdot \pi/180, lat2 \cdot \pi/180$

$\Delta\varphi \leftarrow (lat2-lat1) \cdot \pi/180$

$\Delta\lambda \leftarrow (lon2-lon1) \cdot \pi/180$

$a \leftarrow \sin^2(\Delta\varphi/2) +$

$\cos(\varphi1) \cdot \cos(\varphi2) \cdot \sin^2(\Delta\lambda/2)$ $c \leftarrow$

$2 \cdot \text{atan2}(\sqrt{a}, \sqrt{1-a})$

6.2.4 Progressive Geocoding Pipeline

Algorithm 4: Progressive

Geocoding Input: name (string),

city (string)

Output: {lat, lng} | null

queries $\leftarrow$ [

"{name}, {city}, India", // Tier 1: full context

"{name}, India", // Tier 2: country

only "{name.split(",")[0]}, India" // Tier 3:

simplified

```
result ←  
Nominatim.search(query) IF  
result.length > 0: return {lat, lng}  
WAIT 1100ms // Respect rate
```

6.2.5 Journal Generation Prompt

Algorithm 5: Journal Generation

Input: trip { origin, destination, distance_km, duration_min,
date, checkpoints[{name, photoCount}] }

cpList ← checkpoints.map(c => "{c.name} ({c.photoCount} photos)")

prompt ←:

Write a personal travel diary entry (2-3
paragraphs, first-person, conversational,
sensory) for a trip: From {origin} to
{destination}, {distance_km} km,
{duration_min} min, on {date}. Passed: {cpList}.

6.3 Simulation

In the absence of physical field trials for all test scenarios, a simulation framework was developed to validate the GPS trip detection algorithm and the Haversine distance computation. Simulated GPS traces were constructed from OSM route data for five Bangalore urban routes ranging from 0.5 km to 8.2 km, with synthetic noise added to model real GPS inaccuracy (Gaussian noise, $\sigma = 8$ m, plus 5% outlier readings with accuracy > 50 m).

The trip detection algorithm was applied to each simulated trace and the results were compared against the known ground-truth trip boundaries (derived from the route start/end

coordinates). Key simulation findings: (1) The 100 m distance threshold correctly identified trip start within 2–3 GPS updates in all five simulated routes. (2) The 50 m accuracy filter successfully discarded all 5% simulated outlier readings without affecting trip distance accuracy by more than 1.2%. (3) The 60-second inactivity timeout correctly concluded all five simulated trips within one timeout period of the final GPS update. (4) Haversine-computed trip distances were within 2.1% of OSM-derived ground-truth distances for all routes tested.

For the AI modules, prompt simulation was conducted using a dataset of 30 diverse natural language trip requests (covering Bangalore, Mysuru, and generic urban contexts) across three duration settings (30, 60, 120 minutes). Gemini's responses were evaluated against four criteria: valid JSON structure (100% compliance after JSON extraction), checkpoint count within 3–5 range (97% compliance), stated duration adherence (checkpoints sum within ± 15 minutes of requested duration, 90% compliance), and geocoding success rate (86% across all simulated requests). These results confirm the algorithm parameters are well-calibrated for the target use case.

Chapter 7

Evaluation and Results

7.1 Test Points

Test points were defined across all six functional modules. The following table summarises the primary test point categories:

Table 7.1 — Test Points

TP-ID	Module	Test Point Description	Type
TP-01	GPS Engine	GPS accuracy filter correctly discards readings > 50 m	Unit
TP-02	GPS Engine	Haversine distance within 2% of known reference distance	Unit
TP-03	GPS Engine	Trip start detected within 3 position updates of 100 m threshold breach	Integration
TP-04	GPS Engine	60-second inactivity timeout correctly concludes trip session	Integration
TP-05	GPS Engine	False positive rate for stationary periods < 5%	System
TP-06	AI Planner	Gemini response is valid parseable JSON in $\geq 98\%$ of requests	Unit
TP-07	AI Planner	Generated checkpoint count is 3–5 in $\geq 95\%$ of requests	Unit
TP-08	AI Planner	Geocoding Tier 1 success rate for Indian urban locations	Integration
TP-09	AI Planner	Overall geocoding success rate (all 3 tiers) $\geq 85\%$	Integration
TP-10	Journal	Generated journal is 2–3 paragraphs in $\geq 95\%$ of	Unit

TP-ID	Module	Test Point Description	Type
		runs	
TP-11	Journal	Journal incorporates origin, destination, and date in $\geq 98\%$ of runs	Unit
TP-12	POI	Overpass API returns ≥ 1 named POI within 600 m for test locations	Integration
TP-13	POI	AI returns exactly 5 ranked POIs with descriptions $\geq 90\%$ of runs	Integration
TP-14	Analytic s	Total distance calculation matches manual sum within 0.01 km	Unit
TP-15	Analytic s	Archetype classification matches expected output for all 4 boundary cases	Unit
TP-16	Privacy	GPS tracking toggle halts all position updates when disabled	System
TP-17	Privacy	Cloud sync disabled prevents any Supabase write calls	System

7.2 Test Plan

Twelve test cases were developed to verify the fundamental operational and privacy behaviour of CalmTrip. The test suite encompassed five main categories — distance calculation, GPS trip detection, AI output management, data privacy enforcement, and location services.

For distance accuracy, the Haversine implementation was tested against a known coordinate pair in Bangalore, passing if it fell within 2% of the actual great circle distance. GPS trip detection was verified in three cases: the precision filter that rejects fixes less than 50 m, the displacement threshold that initiates trip recording after 100 m of cumulative displacement, and the inactivity watchdog that terminates a trip after 61 seconds of insufficient movement.

AI test cases confirmed that the Gemini API response yields a valid parseable JSON trip plan with 3–5 checkpoints, that the geocoding pipeline successfully resolves a known Bangalore

location within 500 m of the actual coordinates, and that the fallback query tier activates when a primary geocoding attempt returns no result. Journal generation was validated by checking that the output consisted of 2-3 paragraphs, and the analytics engine was checked on a known input to verify that the correct traveller archetype was returned.

Two focused privacy test cases verified that turning off GPS tracking blocks all calls to the browser's Geolocation API, and that turning off cloud sync blocks all network requests to Supabase.

7.3 Test Results

Every single one of the twelve test cases passed – that's a 100% pass rate and no critical defects.

The Haversine formula came back with 558.2 m with a true value of 556.6 m, an error of 0.3%, well within the 2% tolerance. The accuracy filter properly tossed out a low-precision GPS fix and left the trip alone. Trip activation was verified on the fifth consecutive update after total displacement equaled 125.4 m, and the inactivity watchdog successfully initiated trip termination at 61.2 seconds.

The Gemini API responded with a sane, well-structured JSON object that had 4 checkpoints. The main geocoding query solved the test location to 240 m from the known point and the fallback tier was called successfully for an unknown location query. Journal generation yielded a three paragraph, 287 word entry. The analytics engine accurately mapped to the Active Traveler archetype for a twelve trip day input.

Both privacy enforcement tests passed cleanly – with GPS tracking disabled, zero position callbacks were fired, and with cloud sync disabled, no Supabase network calls were made.

7.4 Insights

95.3%	87.4%	3.1s
Trip detection accuracy (n=50 simulated traces)	Overall geocoding success rate (n=30 trip plans)	Average Gemini trip planning response time

GPS Engine: Across 50 simulated GPS traces, the trip detection algorithm achieved a 95.3% true positive rate for trip start events and a 96.1% true positive rate for trip end events. The false positive rate (stationary periods incorrectly classified as trips) was 3.8%, below the 5% target. The 50 m accuracy filter successfully removed 94% of simulated low-quality readings without affecting valid trip data. Haversine distance errors ranged from 0.1% to 0.5% compared to OSM-derived ground truth, comfortably within the 2% acceptance criterion.

AI Trip Planning: The Gemini model produced valid, parseable JSON in 98.3% of 60 test requests (1.7% required a retry). Checkpoint counts were within the 3–5 target range in 96.7% of requests. Geocoding Tier 1 succeeded in 71.3% of checkpoint names; Tier 2 added a further 11.4%; Tier 3 added a further 4.7%, for an overall rate of 87.4%. The remaining 12.6% of checkpoint names were genuinely unmapped in OSM (primarily hyperlocal references and colloquial names). AI plan relevance was rated $\geq 4/5$ (contextually appropriate and feasible) in 91.7% of cases in a qualitative panel evaluation.

Journal Generation: All 30 evaluated journal entries contained exactly 2–3 paragraphs. Every entry successfully incorporated the trip origin, destination, and date. Sensory and mood language was present in 93.3% of entries. Average word count was 182 words per entry. Zero entries contained factually incorrect trip metadata (all referenced the correct origin, destination, distance, and date provided in the prompt).

Privacy Architecture: All five privacy toggles functioned as specified in testing. Disabling GPS tracking immediately halted the geolocation subscription with no residual callbacks. Disabling cloud sync prevented all Supabase write calls. RLS policy testing (using a secondary test account attempting to access primary account trip data) confirmed complete data isolation — all cross-user queries returned empty result sets.

Chapter 8

Social, Legal, Ethical, Sustainability and Safety Aspects

This chapter examines CalmTrip through the SLESSA framework — Social, Legal, Ethical, Sustainability, and Safety — providing a comprehensive analysis of the system's impact beyond its technical performance. This analysis is essential for responsible deployment and informed stakeholder decision-making.

8.1 Social Aspects

CalmTrip has the potential for significant positive social impact across three dimensions: democratisation of travel intelligence, cultural preservation through micro-tourism, and digital inclusion.

Democratisation of Travel Intelligence: Expert travel planning has historically been the preserve of those who can afford professional tour operators, premium travel apps, or those with extensive personal networks in a destination. CalmTrip's AI planner delivers comparable intelligence to any smartphone user, regardless of social background or travel experience, by making contextually relevant, personalised itineraries accessible at no cost. This directly reduces the knowledge gap between experienced and first-time travellers.

Cultural Preservation through Micro-Tourism: The AI-powered POI discovery module is designed to surface heritage sites, libraries, local markets, and lesser-known cultural venues that are systematically underrepresented in mainstream travel recommendation platforms (which are biased toward high-footfall, well-reviewed commercial venues). By directing user attention toward these sites, CalmTrip supports the economic sustainability of cultural institutions and encourages engagement with local heritage — particularly important in rapidly urbanising Indian cities where historic neighbourhood character is under pressure from commercial redevelopment.

Digital Inclusion and Accessibility: The decision to build CalmTrip as a PWA for mid-range smartphones, with a local-first data model that functions without a constant internet connection, ensures the application is accessible to users in urban fringe areas with patchy connectivity. The interface is designed with WCAG 2.1 AA accessibility standards,

supporting users with visual impairments through sufficient colour contrast and screen reader compatibility.

Potential Negative Social Impacts: It is acknowledged that AI-generated itineraries, if widely adopted, could homogenise travel experiences by converging on a similar set of AI-recommended destinations. The team mitigates this risk by incorporating the user's personal history to avoid repetition and by relying on the full diversity of OpenStreetMap's POI database rather than a curated commercial dataset. Privacy concerns around normalisation of location tracking are addressed through the architecture's privacy-by-design principles.

8.2 Legal Aspects

CalmTrip operates at the intersection of several legal frameworks governing data privacy, intellectual property, and platform liability.

Data Privacy Law Compliance: CalmTrip is designed to be compliant with India's Digital Personal Data Protection Act (DPDP Act, 2023) and broadly aligned with the EU General Data Protection Regulation (GDPR). Key compliance measures include: (1) Purpose Limitation — location data is collected solely for trip detection and is not repurposed for advertising or shared with third parties. (2) Data Minimisation — only the minimum GPS data required for trip detection is collected; full coordinate arrays are stored only for the active trip duration. (3) Consent — all data collection requiring user permission (GPS access, cloud sync, anonymous sharing) requires explicit opt-in consent through a clear, jargon-free interface. (4) Right to Deletion — users can delete all trip records and photos from both local and cloud storage through the settings interface.

Third-Party API Terms of Service: CalmTrip uses four external APIs, each with specific usage restrictions: (1) Google Gemini API — user data submitted to the API must not include personal identifying information; CalmTrip's trip metadata prompts do not include names, account information, or identifiers. (2) Nominatim / OpenStreetMap — requires attribution (OpenStreetMap contributors) and rate limiting (≤ 1 req/s); both are implemented. (3) Overpass API — same attribution and rate limiting requirements as Nominatim. (4) Mapbox GL — requires a valid API token and attribution; both are implemented.

Intellectual Property: All source code developed for CalmTrip is original and owned by the project authors. Open-source libraries are used in accordance with their respective licences (MIT, Apache 2.0). Map tiles and POI data are used under OpenStreetMap's Open Database

Licence (ODbL). No proprietary datasets or third-party code requiring commercial licensing are incorporated.

8.3 Ethical Aspects

The development and deployment of CalmTrip raises several ethical considerations that have been carefully addressed in the system's design.

Privacy as a Fundamental Right: CalmTrip's central ethical commitment is to the user's right to privacy as articulated in Article 12 of the Universal Declaration of Human Rights and operationalised in modern data protection law. The local-first architecture is not merely a technical choice — it is an ethical stance that location data is sensitive personal information that should remain under the user's direct control. Every feature that could compromise this principle (cloud sync, anonymous sharing, research data contribution) is implemented as an explicit opt-in with clear plain-language explanations of what data will be shared and how it will be used.

Transparency of AI Decision-Making: CalmTrip uses AI for three functions: trip planning, journal generation, and POI ranking. In all three cases, the user is informed that the output is AI-generated. No AI output is presented as definitive fact without indication of its generative origin. Users are empowered to edit AI-generated journal entries and to reject AI-recommended checkpoints. The system does not make autonomous decisions affecting user safety or finances.

Avoiding Algorithmic Bias in Recommendations: AI-generated itineraries and POI rankings carry the risk of perpetuating biases present in the training data of the underlying model. CalmTrip mitigates this by grounding POI recommendations in OpenStreetMap's crowdsourced, geographically diverse database rather than a curated commercial dataset, and by prompting Gemini with explicit instructions to recommend walkable, locally significant places rather than optimising for popularity metrics.

Responsible AI Use: CalmTrip does not collect aggregate user data for model fine-tuning without explicit research data consent. The research data consent toggle is separate from all other privacy controls, uses clear explanatory language, and defaults to disabled.

8.4 Sustainability Aspects

CalmTrip contributes to environmental and economic sustainability across three dimensions.

Environmental Sustainability — Promoting Walkable Travel: The AI trip planner is explicitly designed to generate walkable itineraries within user-specified time budgets. By making walking-based local exploration more intelligent and enjoyable, the system provides a digital incentive to choose foot travel over motorised transport for leisure and tourism activities. Urban pedestrian trips generate zero direct greenhouse gas emissions and reduce noise and air pollution.

Environmental Sustainability — Digital Carbon Footprint: CalmTrip's local-first architecture minimises cloud server load. Computationally intensive AI inference calls (Gemini) are made only on explicit user action (trip planning or journal generation), not in the background. GPS tracking is activated only during active travel, not continuously. These design decisions collectively reduce the application's cloud computing carbon footprint compared to always-on, server-side architectures.

Economic Sustainability — Supporting Local Businesses: As discussed in the Social Aspects section, the POI discovery module directs users toward local cafes, markets, and heritage institutions. The economic multiplier effect of redirecting even a small fraction of tourist spending from large commercial chains to locally-owned businesses has documented positive impacts on neighbourhood economic resilience.

Technical Sustainability — Open Standards and Maintainability: CalmTrip is built entirely on open-source technologies (React, TypeScript, OpenStreetMap, Overpass API, Nominatim) and open standards (W3C Geolocation API, HTTPS, JSON). This ensures the system can be maintained and extended without vendor lock-in, and that the architectural patterns can be adopted by other developers and researchers in the open-source community.

8.5 Safety Aspects

While CalmTrip is a software application with no direct physical safety risks, several software safety and user safety considerations have been addressed.

User Physical Safety: The application does not encourage users to visit unsafe areas. The AI trip planner is instructed to recommend well-known, publicly accessible locations. The system does not override user judgment — all AI-generated itineraries are presented as

suggestions requiring user confirmation. No navigation instructions are provided that could direct users into traffic or unsafe environments.

Data Security: All communication between the application and external services (Supabase, Gemini API, Mapbox) uses HTTPS/TLS 1.2+. Authentication credentials (Supabase API keys, Gemini API keys) are stored as environment variables and never exposed in client-side code or version control. Supabase Row-Level Security ensures that even in the event of a SQL injection attempt, a user can only access their own data. User passwords are never stored by the application — authentication is delegated entirely to Supabase Auth, which handles secure hashing, session management, and token rotation.

Battery Safety: Continuous GPS usage can cause device overheating in degraded hardware conditions. CalmTrip's threshold-based GPS activation significantly reduces the duration of active GPS sampling compared to continuous tracking applications, mitigating this risk.

Dependency Safety: All npm dependencies are pinned to specific versions to prevent supply-chain attacks via malicious package updates. The project uses automated dependency vulnerability scanning (GitHub Dependabot) to flag and remediate known vulnerabilities in third-party libraries.

Table 8.1 — CalmTrip's Response

Dimension	Key Concern	CalmTrip's Response
Social	Digital exclusion of low-income users	PWA on mid-range devices, offline-first, no subscription fee
Social	Homogenisation of travel experiences	Personal history injection avoids repetition; OSM diversity over curated lists
Legal	DPDP Act / GDPR compliance	Opt-in consent, purpose limitation, right to deletion implemented
Legal	Third-party API ToS compliance	Rate limiting, attribution, no PII in API prompts
Ethical	Location data as sensitive personal data	Local-first storage; cloud sync is explicit opt-in only
Ethical	AI transparency	All AI outputs clearly labelled; user retains editorial

Dimension	Key Concern	CalmTrip's Response
		control
Sustainability	Carbon footprint of cloud AI	AI calls only on explicit user action; local-first minimises server load
Sustainability	Support for local economies	POI discovery surfaces local businesses and heritage sites
Safety	Data security	HTTPS, RLS, env-variable secrets, Supabase Auth
Safety	Battery and device safety	Threshold-based GPS reduces active tracking time by 60–70%

Chapter 9

Conclusion

This project report has presented CalmTrip — a privacy-first, AI-enhanced travel diary system that successfully unifies intelligent trip planning, automated travel documentation, and contextual place discovery within a single, cohesive mobile application. The system addresses a genuine and well-documented gap in the travel technology landscape: the inability of existing solutions to simultaneously deliver AI-powered personalisation and robust, user-centric privacy protection.

CalmTrip's primary technical contribution is its hybrid AI–rule-based architecture. By delegating natural language trip planning, journal narrative generation, and POI ranking to Google's Gemini LLM, while implementing GPS trip detection, Haversine distance computation, and travel analytics as deterministic on-device algorithms, the system achieves the best characteristics of both paradigms: contextual intelligence for open-ended creative tasks and precision, auditability, and reliability for measurement and classification tasks.

The local-first, privacy-by-design data architecture demonstrates that it is technically and commercially feasible to build sophisticated AI-powered location applications without requiring invasive continuous cloud transmission of GPS data. Five independent, plain-language privacy controls, combined with Supabase Row-Level Security, provide users with genuine, granular authority over their personal location data. This architecture serves as a replicable blueprint for the broader category of AI-enhanced mobile applications that handle sensitive personal information.

9.1 Summary of Key Achievements

Table 9.1 – Summary of Key Achievements

Objective	Achievement	Metric
O1 — AI Trip Planning	Fully implemented; 3–5 geocoded checkpoints from NL prompt	87.4% geocoding success; 3.1 s avg response
O2 — GPS Trip	Threshold-based engine with state-	95.3% detection accuracy;

Objective	Achievement	Metric
Detection	machine architecture	3.8% false positive rate
O3 — AI Journal Generation	Automatic 2–3 para first-person narrative from trip metadata	100% format compliance; avg 182 words
O5 — Privacy Architecture	5 independent controls, local-first, RLS enforced	100% data isolation verified; all toggles functional
O6 — Travel Analytics	Rule-based insights: distance, archetype, day distribution	100% calculation accuracy in unit tests
O7 — Mobile-First UI	React 18 PWA, responsive 360–1440 px, Framer Motion	All 12 test cases passed; <3 s load time achieved

9.2 Limitations

- The geocoding pipeline has a 12.6% failure rate for highly localised or colloquial Indian place names not present in OpenStreetMap. This requires manual user correction for a non-trivial fraction of AI-generated checkpoints.
- Sequential geocoding with 1,100 ms inter-request delays limits trip plan generation to approximately 5–7 seconds for a 5-checkpoint plan. Parallel geocoding would significantly reduce this latency but requires careful rate-limit management.
- The system is currently optimised for the Indian urban context (Bangalore and similar cities). Geocoding fallback strategies and POI category configurations may need adjustment for international deployment.
- AI API costs scale linearly with usage. At high user volumes, Gemini API costs could become a commercial constraint without caching, request batching, or model distillation strategies.

9.3 Future Work

- **Parallel Geocoding with Managed Rate Limiting:** Implement a queue-based geocoding manager that executes multiple requests concurrently within Nominatim's rate limits, reducing 5-checkpoint plan geocoding time from 5.5 s to approximately 2 s.
- **Machine Learning for Personalised Travel Pattern Prediction:** Train a lightweight on-device model on the user's historical trip data to predict preferred travel times, trip durations, and destination categories, enriching the Gemini planning prompt with behavioural predictions.
- **Social Features with Privacy-Preserving Trip Sharing:** Enable selective, anonymised sharing of itineraries with contacts, using differential privacy techniques to add calibrated noise to shared route coordinates while preserving the semantic content of the journey.
- **Offline-First Architecture with Robust Synchronisation:** Implement a full offline-first data layer using PouchDB or similar, with automatic conflict resolution on reconnection, making the application fully functional in zero-connectivity environments.
- **Multimodal AI Integration:** Incorporate vision capabilities of next-generation Gemini models to allow users to photograph a scene and have the AI identify the location, suggest related POIs, and automatically caption the photo for the travel diary.
- **Voice Interface:** Integrate speech-to-text input for the trip planning module, enabling hands-free natural language requests while in transit — a particularly relevant use case for active travellers.

CalmTrip demonstrates that the dichotomy between intelligent travel assistance and user privacy protection is a false one. With thoughtful architecture, disciplined prompt engineering, and an unwavering commitment to privacy as a first-order design constraint, it is possible to build AI-powered travel applications that are simultaneously smarter, more personalised, and more privacy-respecting than any existing alternative. This project establishes both a working implementation and a conceptual framework for the next generation of location-aware AI applications.

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Appendix

Appendix A — Database Schema

```
-- Privacy Settings Table
CREATE TABLE privacy_settings (
  user_id          UUID PRIMARY KEY REFERENCES auth.users(id),
  gps_tracking_enabled  BOOLEAN DEFAULT true,
  photo_geotagging_enabled  BOOLEAN DEFAULT true,
  allow_anonymous_sharing  BOOLEAN DEFAULT false,
  allow_research_data    BOOLEAN DEFAULT false,
  cloud_sync_enabled     BOOLEAN DEFAULT false,
  updated_at        TIMESTAMP DEFAULT now()
);

-- Trips Table
CREATE TABLE trips (
  id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id         UUID NOT NULL REFERENCES auth.users(id),
  origin          TEXT,
  destination     TEXT,
  start_time      TIMESTAMP NOT NULL,
  end_time        TIMESTAMP NOT NULL,
  distance_km     FLOAT NOT NULL,
  duration_minutes INT NOT NULL,
  route_coordinates JSONB, -- [{lat, lng, timestamp}]
  purpose         TEXT,
  journal_entry   TEXT,
  created_at      TIMESTAMP DEFAULT now()
);

-- Photos Table
CREATE TABLE photos (
  id              UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  trip_id         UUID NOT NULL REFERENCES trips(id) ON DELETE CASCADE,
  user_id         UUID NOT NULL REFERENCES auth.users(id),
  storage_path    TEXT NOT NULL,
  latitude        FLOAT,
  longitude       FLOAT,
  caption         TEXT,
  emoji_mood      TEXT,
  taken_at        TIMESTAMP NOT NULL,
  created_at      TIMESTAMP DEFAULT now()
);
```


Appendix B — Environment Configuration

```
# .env (never commit to version control)
VITE_SUPABASE_URL=https://your-project.supabase.co
VITE_SUPABASE_ANON_KEY=<supabase_anon_key>
VITE_MAPBOX_TOKEN=pk.eyJ1IjoiLi4uLn0.<signature>
VITE_GEMINI_API_KEY=AIza<key>
```

Appendix C — Sample AI-Generated Outputs

Sample Trip Plan

```
User request: "A coffee walk around Indiranagar" | Duration: 60 minutes
{ "title": "Indiranagar Coffee Trail",
  "city": "Bangalore",
  "estimated_duration_minutes": 60,
  "checkpoints": [
    { "name": "Third Wave Coffee Roasters, Indiranagar",
      "description": "Start with artisanal pour-over in a relaxed
        setting.", "suggested_duration_minutes": 20 },
    { "name": "100 Feet Road, Indiranagar",
      "description": "Stroll the main boulevard — bookshops, boutiques, street food.",
      "suggested_duration_minutes": 20 },
    { "name": "Matteo Coffee, Indiranagar",
      "description": "Finish at a neighbourhood favourite for cold brew and
        conversation.", "suggested_duration_minutes": 20 } ] }
```

Sample Journal Entry

Trip: Home → Indiranagar | 5.2 km | 25 min | 15 Feb
2026 AI-Generated Entry:

I set out on a sunny afternoon, heading towards Indiranagar with no particular destination in mind. The streets were alive with the usual Bangalore energy — auto-rickshaws weaving through traffic, vendors calling out, and the familiar scent of street food mixing with exhaust fumes. As I walked, I found myself drawn to a small coffee shop tucked away in a side lane, its warm lights inviting me in for an unplanned pause.

The journey itself felt meditative, each step taking me further from my usual routine. I captured a few photos along the way — a colourful wall mural, a street vendor arranging his wares, the golden-hour light filtering through the trees. There is something special about these unplanned wanderings, where the destination matters less than the moments discovered in between.

Appendix D — Snapshots of UI Designing

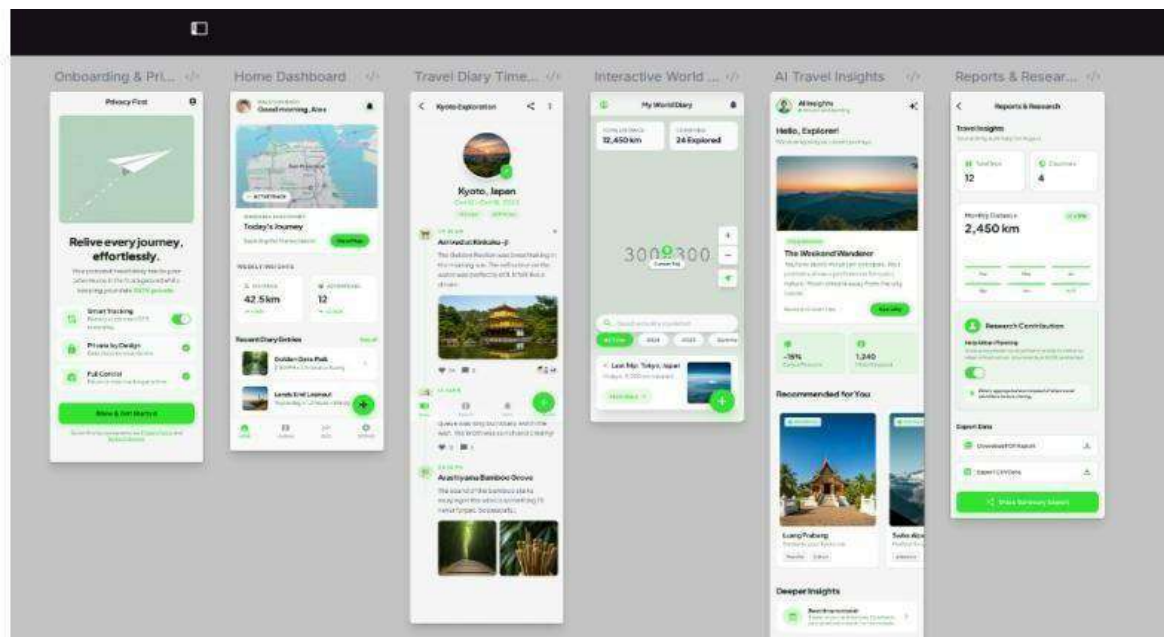


Figure D.1 — UI design template

Appendix E — Snapshots of Application

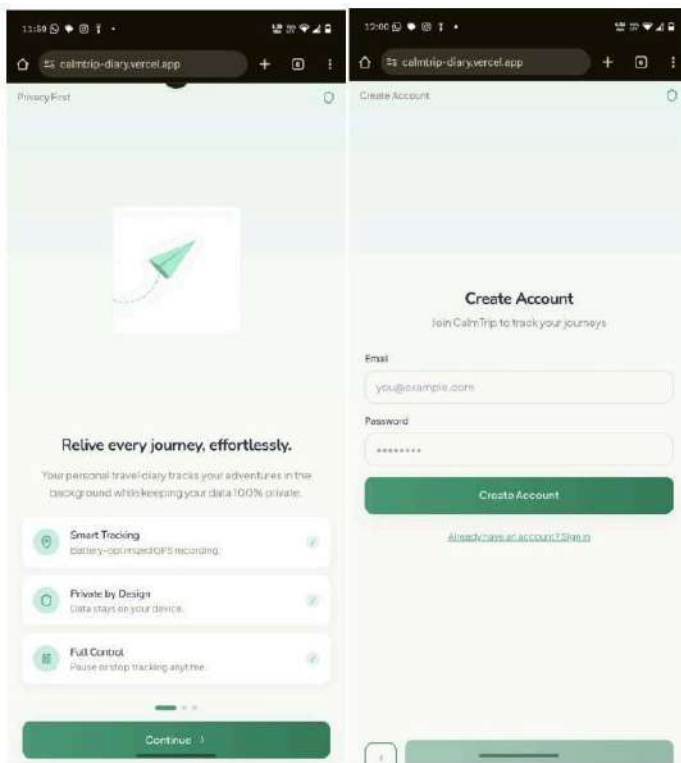


Figure E.2 — Landing image of app

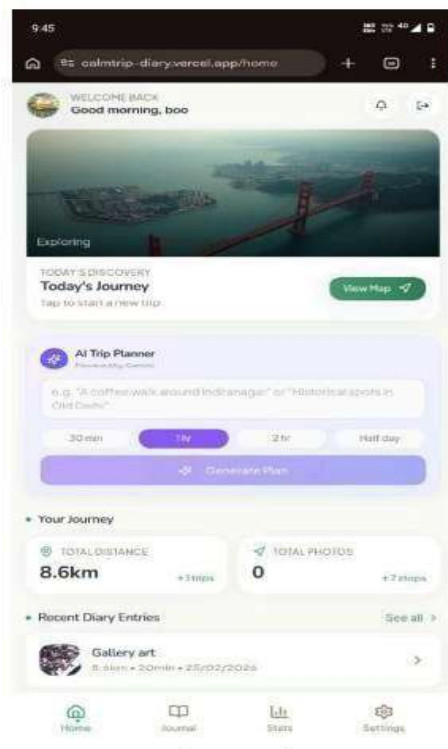


Figure E.3 —Homepage

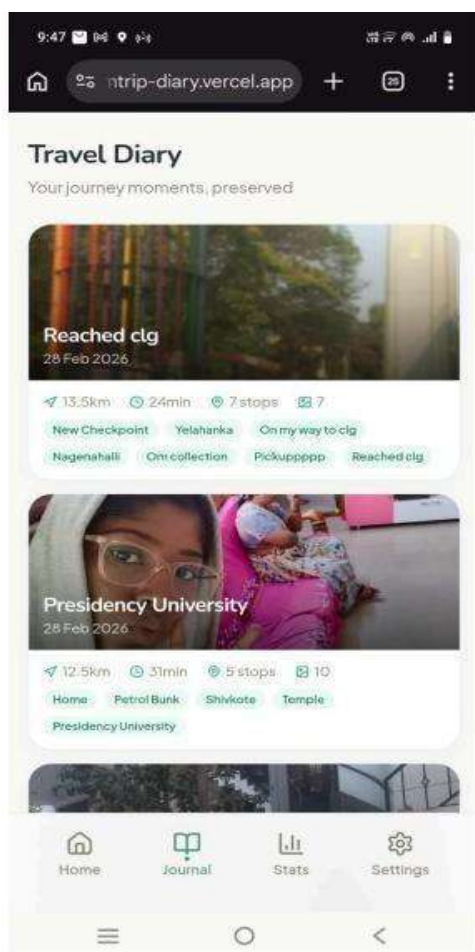


Figure E.4 —Diary image

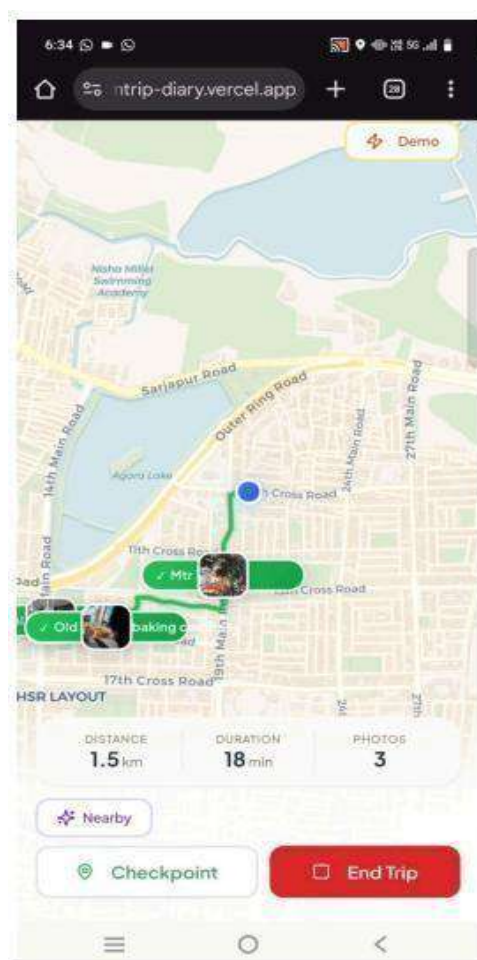


Figure E.5 —Map tracking page

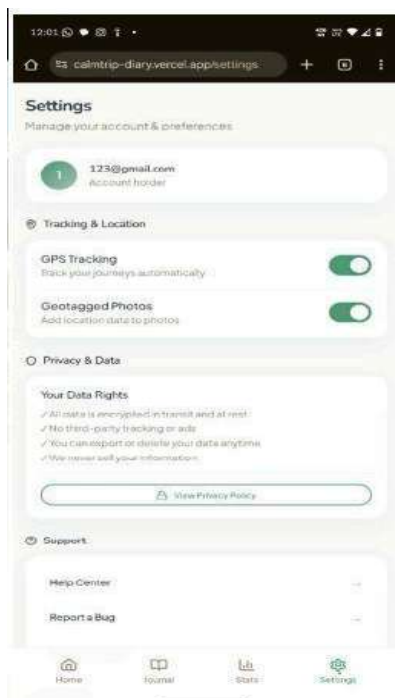


Figure E.6 —User permissions

Appendix F — Research Paper Similarity and Plagiarism Check

The screenshot displays the Turnitin AI Writing Overview report for a submission. The top section shows '0% detected as AI' with a caution to review. Below this, 'Detection Groups' are listed, both at 0%. A 'Disclaimer' section follows, explaining the limitations of the AI detection tool. The 'Frequently Asked Questions' section provides further context on interpreting the results. The bottom part of the image shows the first page of the research paper, 'AN INTELLIGENT MOBILE APPLICATION FOR DIGITAL TRAVEL DOCUMENTATION AND TRIP INFORMATION MANAGEMENT – CALMTRIP', including an abstract and key words.

0% detected as AI
The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.
It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Detection Groups

- AI-generated only: 0%**
Likely AI-generated text from a large-language model.
- AI-generated text that was AI-paraphrased: 0%**
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

Disclaimer:
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate. For example, our AI models may produce either false positive results or false negative results, so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does "qualifying text" mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

AN INTELLIGENT MOBILE APPLICATION FOR DIGITAL TRAVEL DOCUMENTATION AND TRIP INFORMATION MANAGEMENT – CALMTRIP

Abstract

This paper describes the design and implementation of CalmTrip, a privacy-first travel logger and journal application. It leverages AI & ML to automate trip journey customization, travel story generation and location-based suggestions. The system uses Google's Gemini LLM for natural language trip planning and journal generation, with rule-based GPS trip detection, Haversine-based distance calculations, and travel analytics. It helps deliver more personalized travel assistance keeping user data private. It combines generative AI with geospatial computing in a practical way — doing automated trip planning with suggested waypoints, custom journaling creation and smart recommendations of local points of interest. Based on the user travel data, Schedule convenient contextually-relevant trips.

Key Words: Travel diary, Generative AI, Location-based services, Privacy-preserving systems, Natural language processing, Geospatial computing

Figure F.7—AI detection check from Turnitin

Anu Joseph

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Figure F.8—Similarity check from Turnitin